

Compte Rendu Complet de TP ERP

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L2CS02

Acknowledgments

We want to express our sincere gratitude to ***Dr. Zakia ZOUAGHIA*** *for her valuable guidance and support throughout this project, which greatly enhanced our understanding of ERP systems*

Table of Contents

Part I.

I. TP1 1
II. TP2 3
III. TP3 4

Part II. Mini-Project TP ERP

I. Project Introduction 11
II. Introducing Intoy 11
III. Activity 1: Company setup 12
IV. Activity 2: User Management 15
V. Activity 3: Module Installation 17
VI. Activity 4: Partner Management 18
VII. Activity 5: Product Management 20
VIII. Activity 6: Inventory & Warehouse Management 25
IX. Activity 7: Purchasing Workflow 28
X. Activity 8: Sales & Inventory Audit 31
XI. Activity 9: Payment Collection 35
XII. Conception with 2 models BPMN: 36
XIII. Conclusion 37

Part I.

I. TP1

Exercice 1:

1) - An Information System (IS) is an organized set of people, data, procedures, and tools that allow information to be collected, processed, stored, and disseminated within an organization.

- A Computer System: This is the technical part of an information system and includes computers, software, databases, and companies.

2) A business process (processus métier) represents a series of organized activities aimed at producing added value and responding to a specific objective within the organization. Examples include sales, recruitment, and production processes.

3) Business processes define what an organization does, and information systems support and automate these processes, contributing to improved coordination between departments, faster business execution, and smoother operations.

4) Examples of ERP systems:

BAGE, odoo, Microsoft Dynamics, Oracle ERP, SAP, ERPNext

5) Examples of the most important ERP modules:

- Sales and Purchasing Management
- Inventory Management
- Accounting and Finance
- Human Resources

- Production

- Customer Relationship Management (CRM)

- Maintenance

- Marketing

6) Odoo Community: Free and open source, it supports basic functions but lacks official support and advanced applications.

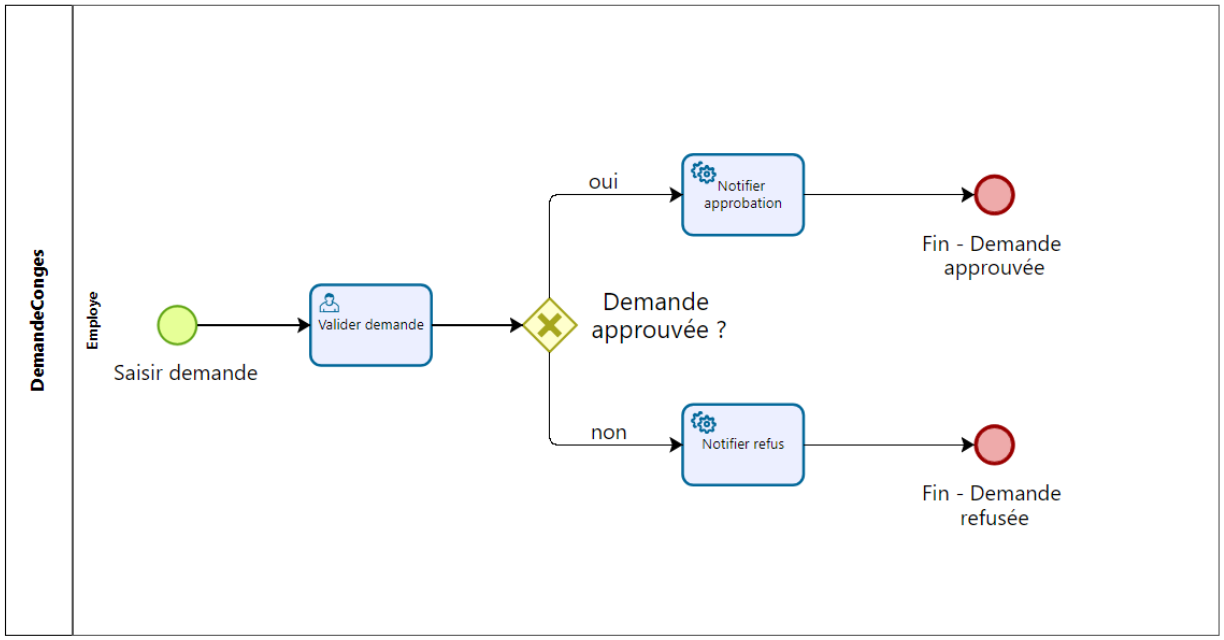
Odoo Enterprise: Paid and closed source, it offers advanced functions, official support, and advanced applications.

7) Benefits of an ERP system for an organization:

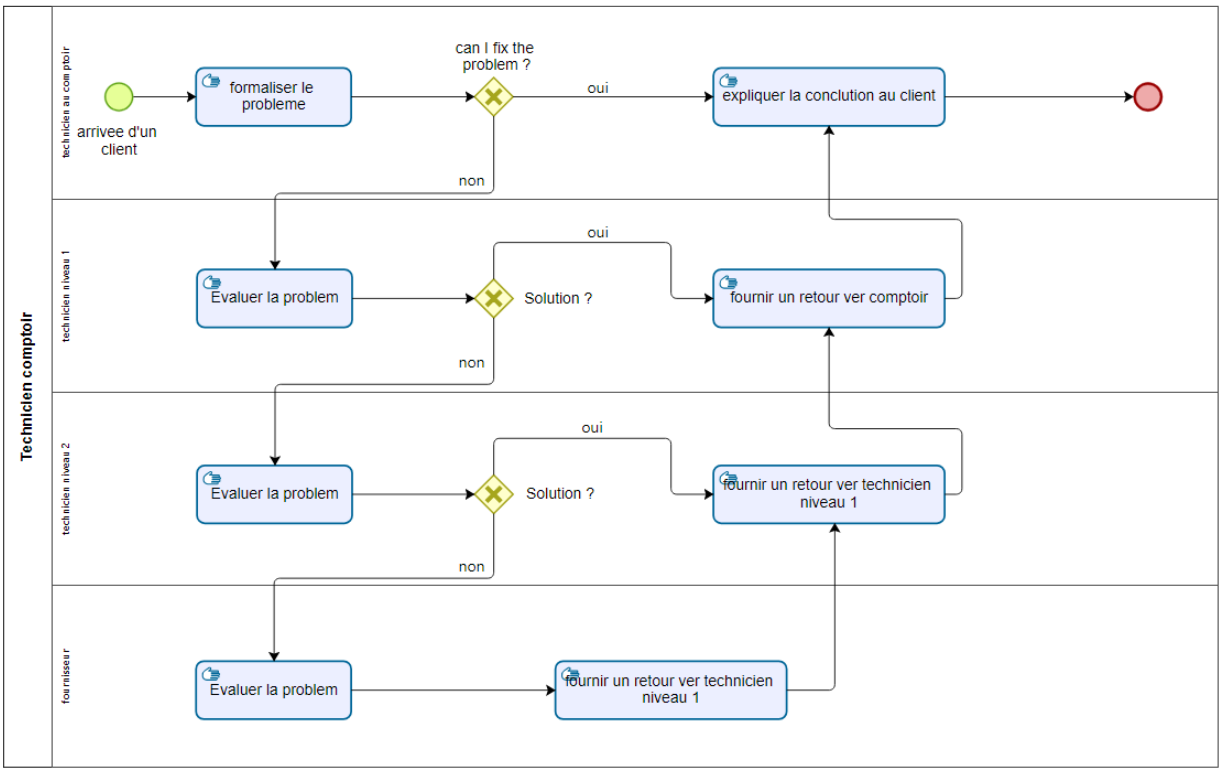
1. Data standardization and integration
2. Improved productivity
3. Reduced errors
4. Support for decision-making
5. Process automation
6. Time and cost savings
7. Improved coordination between different departments

II. TP2

Exercice 1: Modélisation d'un processus simple.

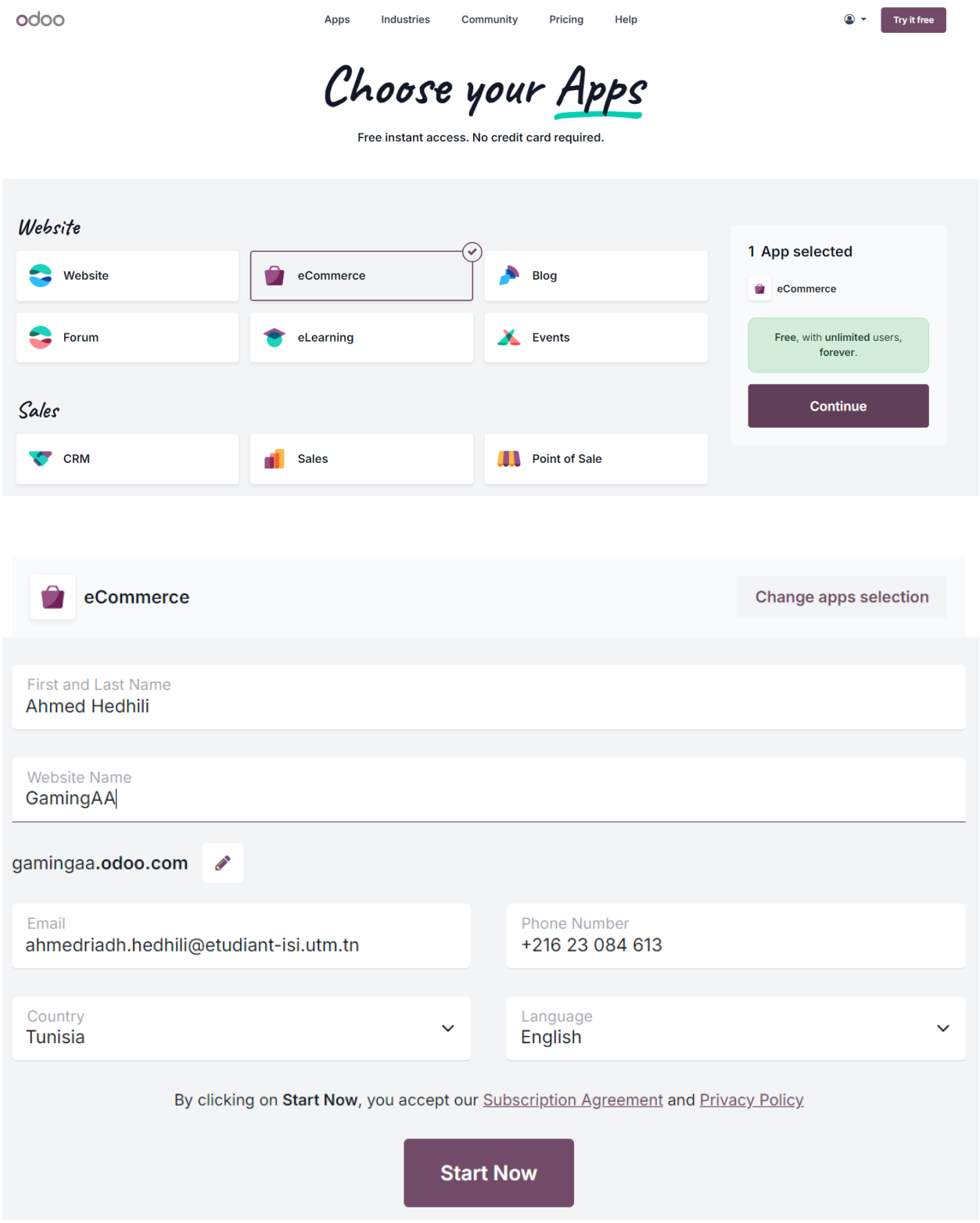


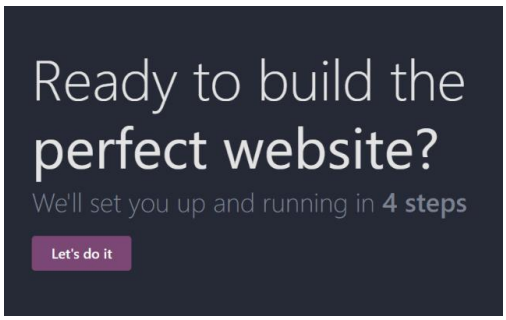
Exercice 2: Modélisation d'un processus de gestion des incidents.

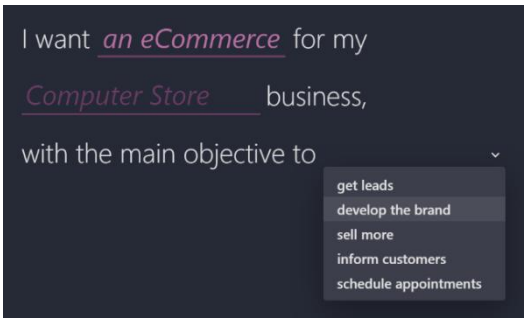


III. TP3

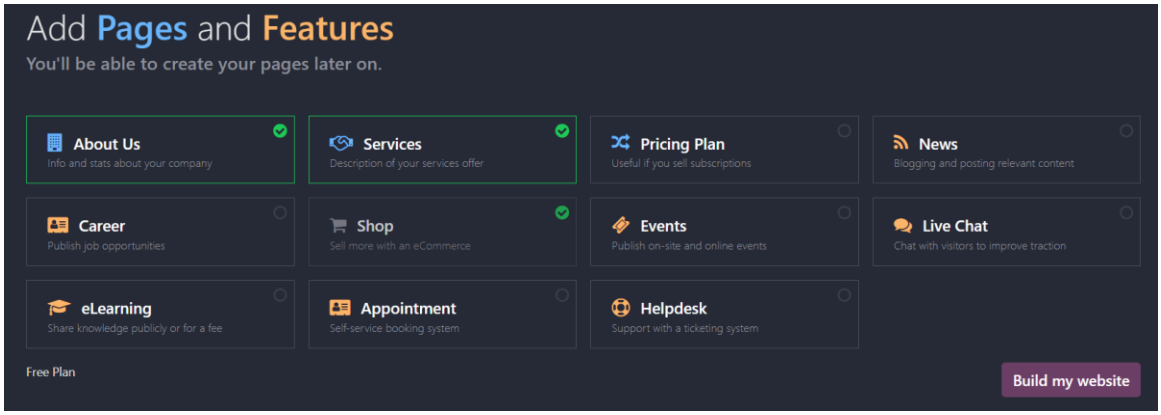
Game purchase website "GamingAA"

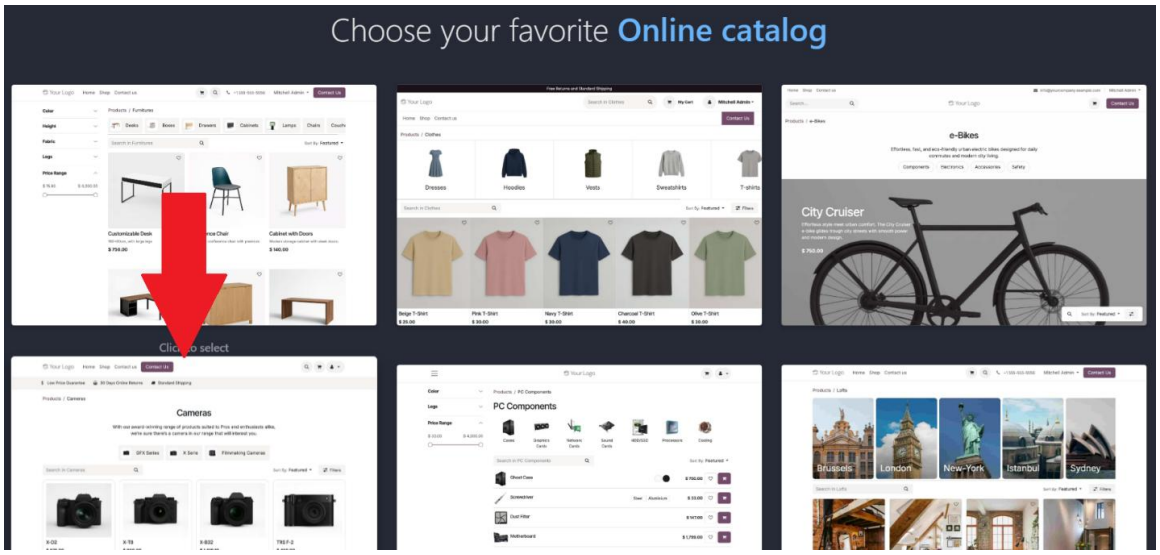


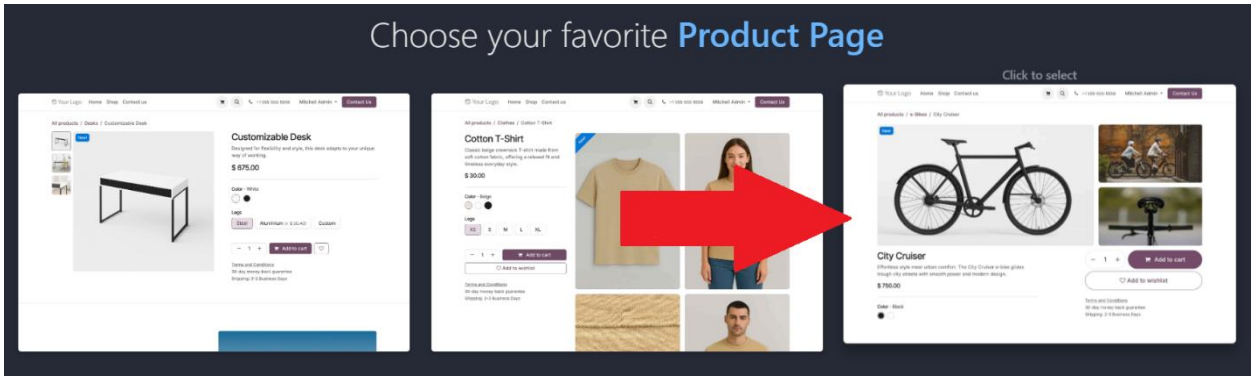

Ready to build the
perfect website?
We'll set you up and running in 4 steps
Let's do it

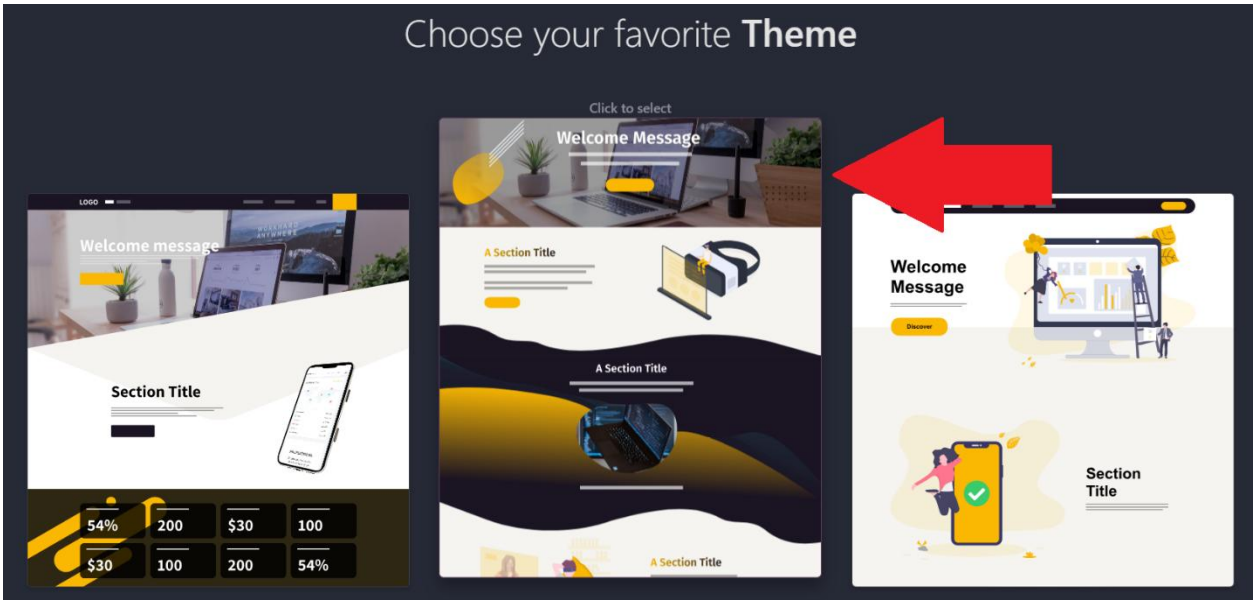

I want an eCommerce for my
Computer Store business,
with the main objective to
get leads
develop the brand
sell more
inform customers
schedule appointments

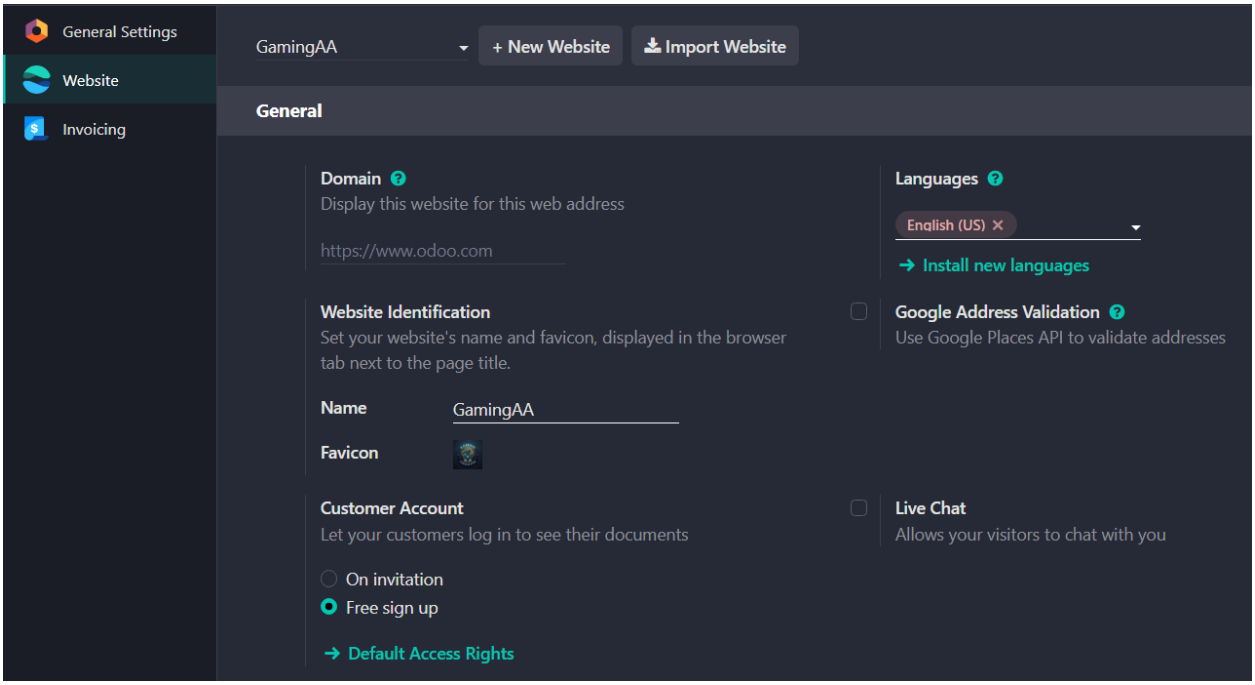

Choose a Color Palette
Detect from Logo
OR
Upload

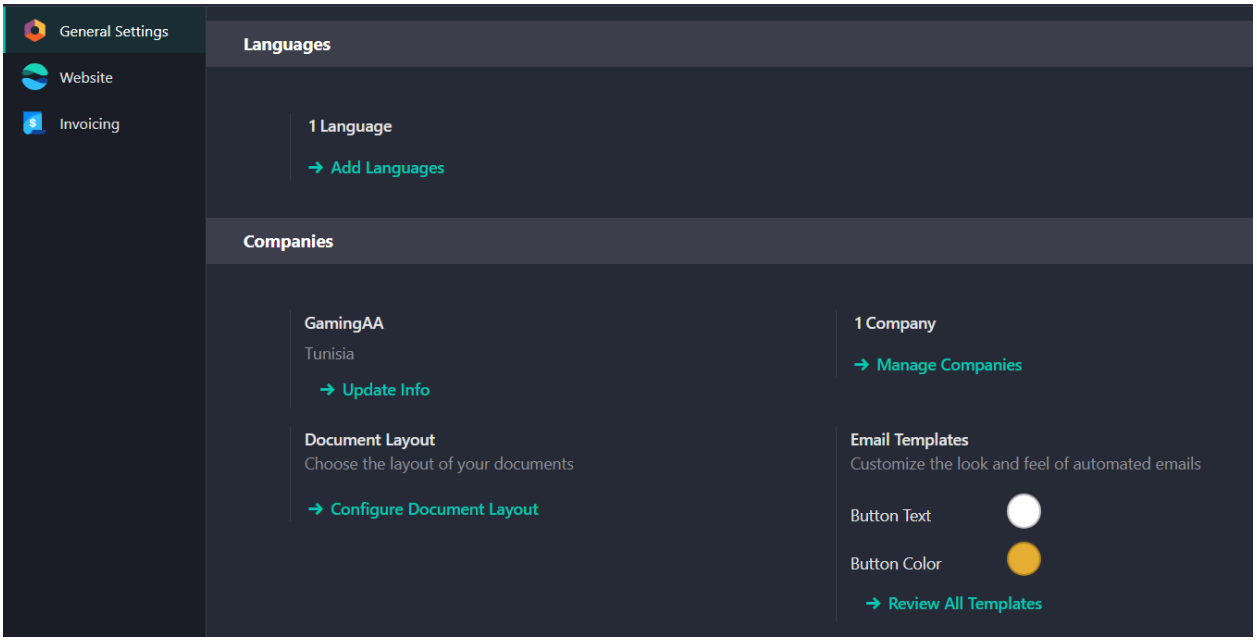

Add Pages and Features
You'll be able to create your pages later on.
About Us
Info and stats about your company
Services
Description of your services offer
Pricing Plan
Useful if you sell subscriptions
News
Blogging and posting relevant content
Career
Publish job opportunities
Shop
Sell more with an eCommerce
Events
Publish on-site and online events
Live Chat
Chat with visitors to improve traction
eLearning
Share knowledge publicly or for a fee
Appointment
Self-service booking system
Helpdesk
Support with a ticketing system
Free Plan
Build my website

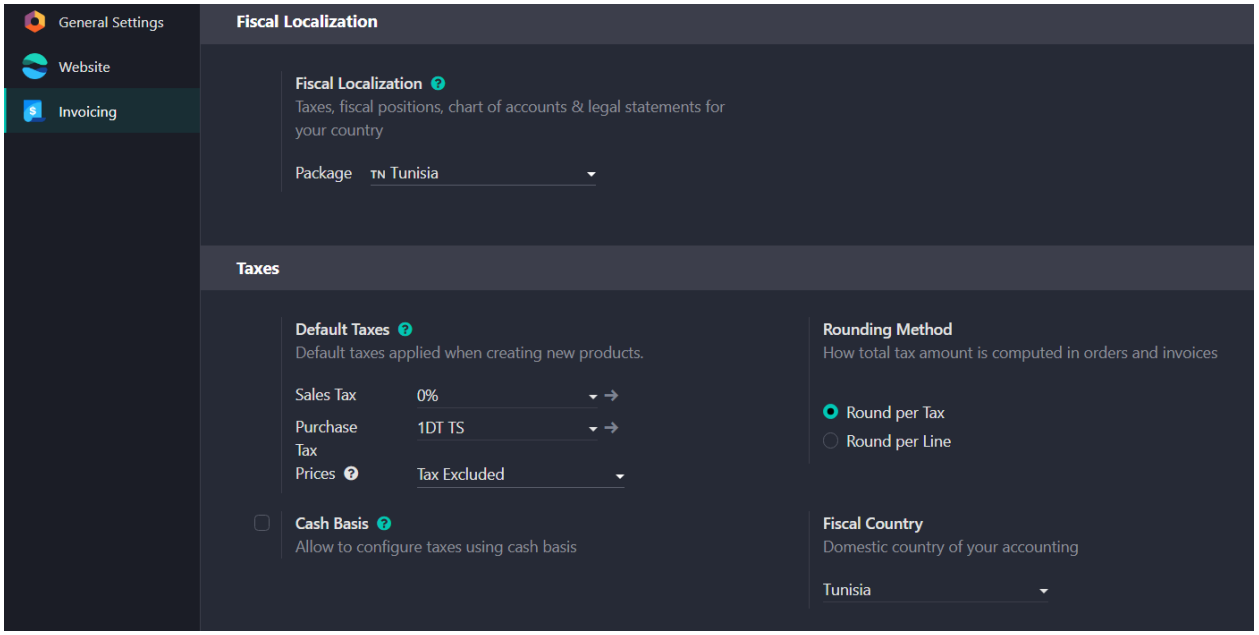

Choose your favorite Online catalog
Cameras
PC Components
e-Bikes
City Cruiser
Brussels
London
New-York
Istanbul
Sydney

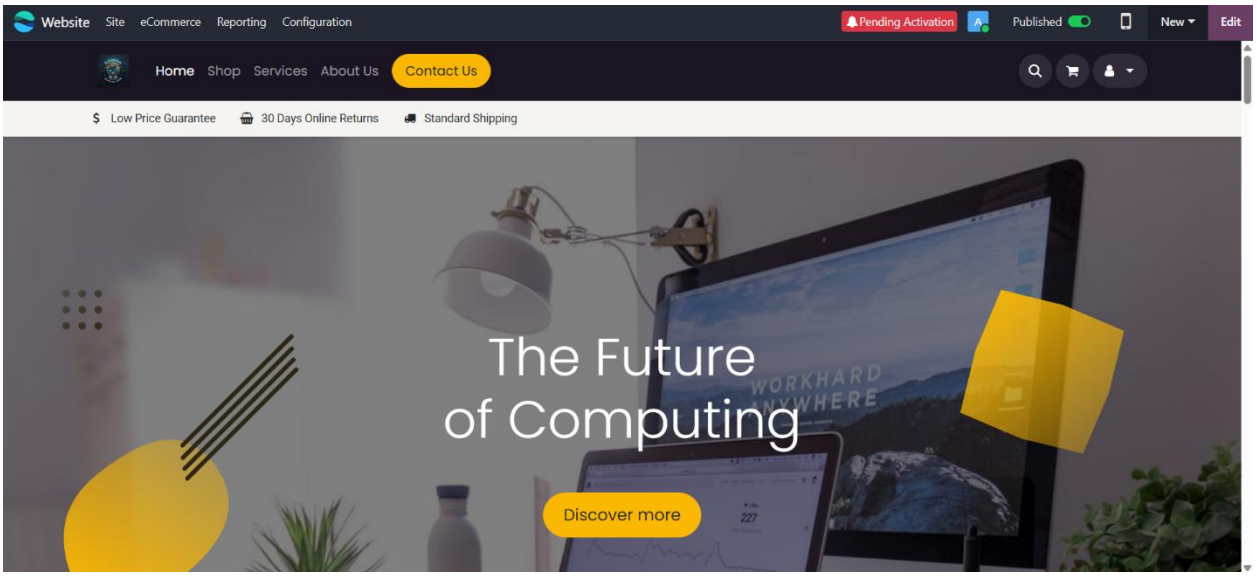

Choose your favorite Product Page
Click to select
Customizable Desk
Cotton T-Shirt
City Cruiser

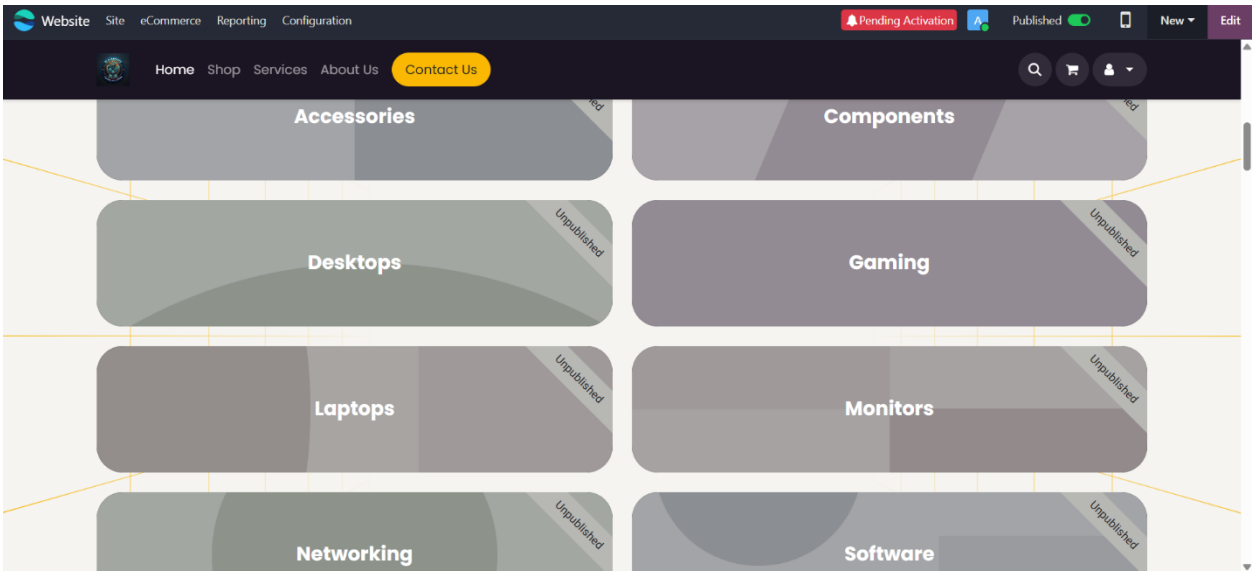

Choose your favorite Theme
Click to select
Welcome Message
Welcome message
Section Title
A Section Title
Welcome Message
Section Title
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200
$30
100
$30
100
200
54%

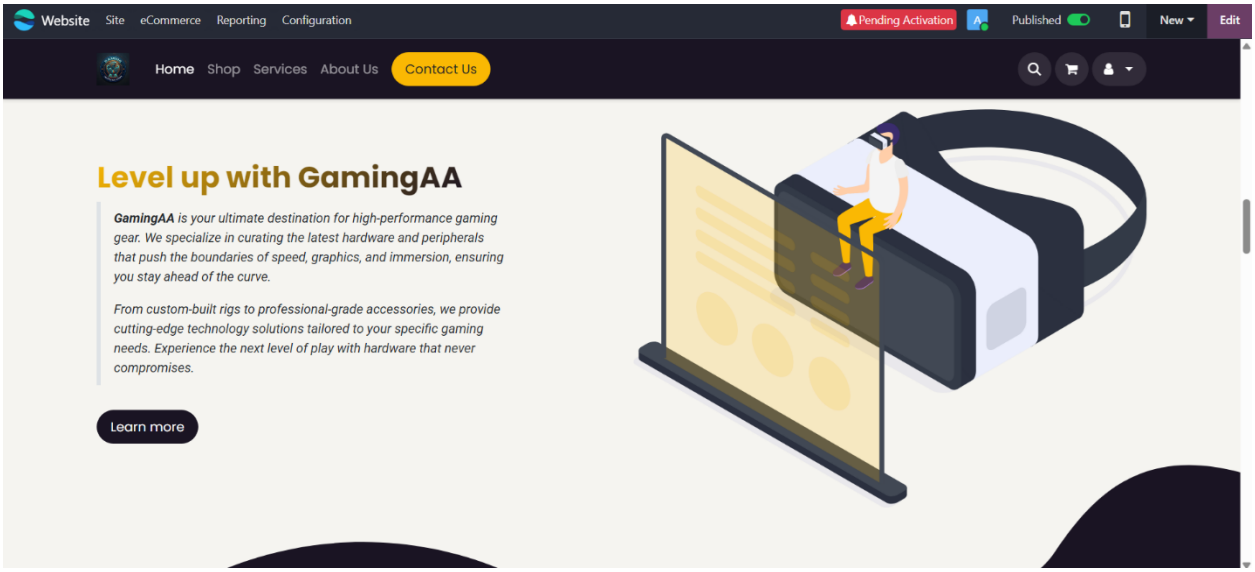

General Settings
Website
Invoicing
GamingAA
+ New Website
Import Website
General
Domain
Display this website for this web address
https://www.odoo.com
Languages
English (US)
Install new languages
Website Identification
Set your website's name and favicon, displayed in the browser tab next to the page title.
Name
GamingAA
Favicon
Google Address Validation
Use Google Places API to validate addresses
Customer Account
Let your customers log in to see their documents
On invitation
Free sign up
Default Access Rights
Live Chat
Allows your visitors to chat with you

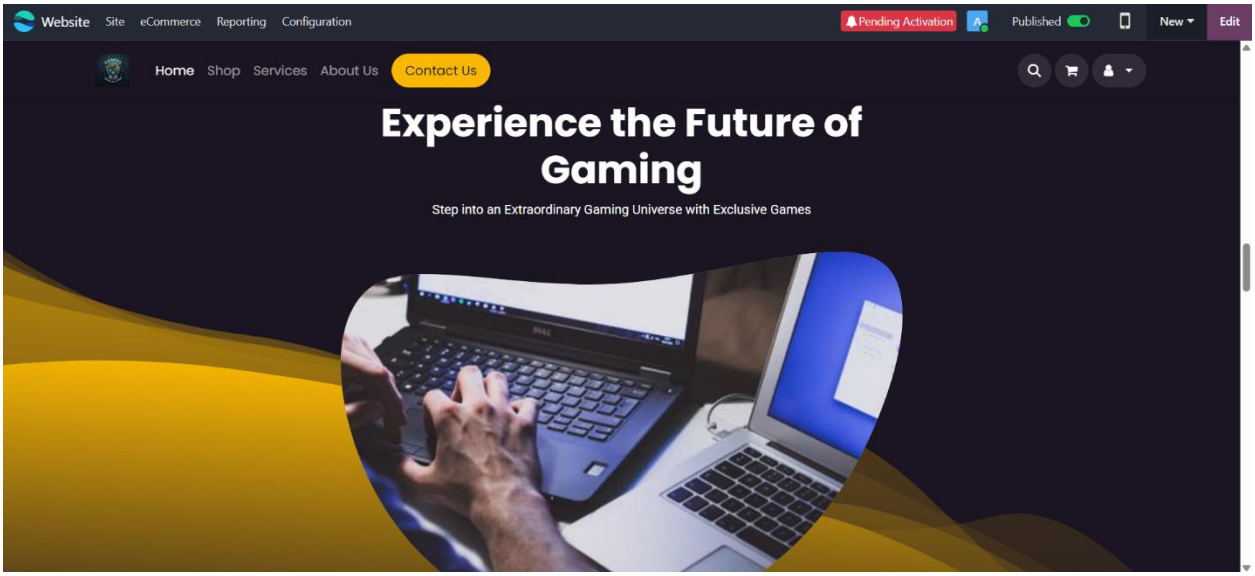

General Settings
Website
Invoicing
Languages
1 Language
Add Languages
Companies
GamingAA
Tunisia
Update Info
1 Company
Manage Companies
Document Layout
Choose the layout of your documents
Configure Document Layout
Email Templates
Customize the look and feel of automated emails
Button Text
Button Color
Review All Templates

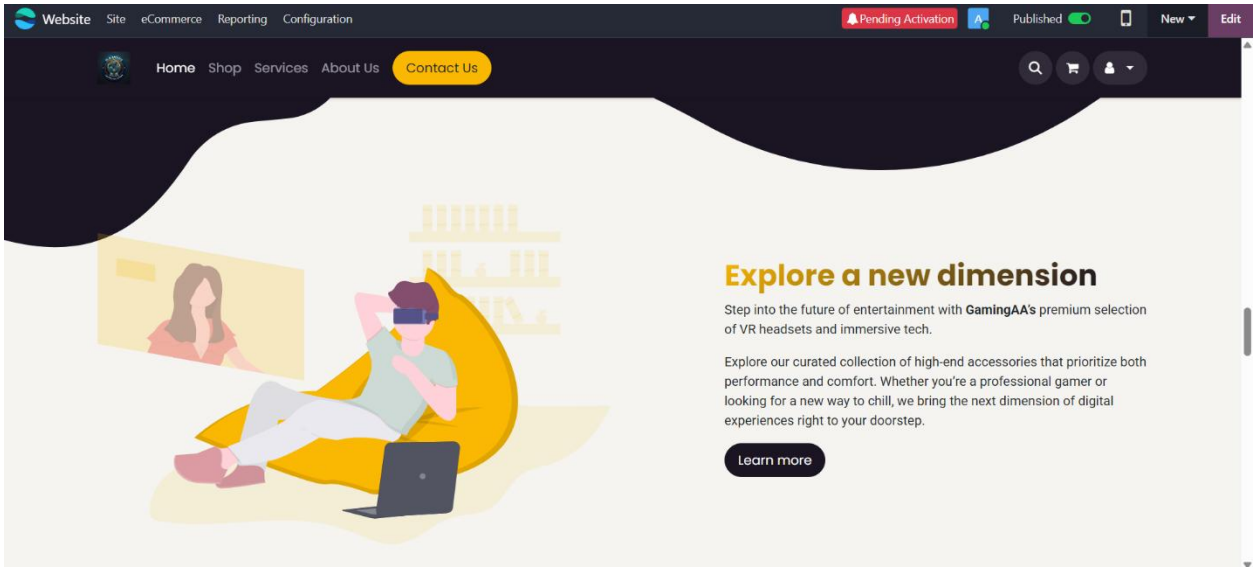

General Settings
Website
Invoicing
Fiscal Localization
Fiscal Localization
Taxes, fiscal positions, chart of accounts & legal statements for your country
Package
TN Tunisia
Taxes
Default Taxes
Default taxes applied when creating new products.
Sales Tax
0%
Purchase Tax
1DT TS
Prices
Tax Excluded
Cash Basis
Allow to configure taxes using cash basis
Rounding Method
How total tax amount is computed in orders and invoices
Round per Tax
Round per Line
Fiscal Country
Domestic country of your accounting
Tunisia

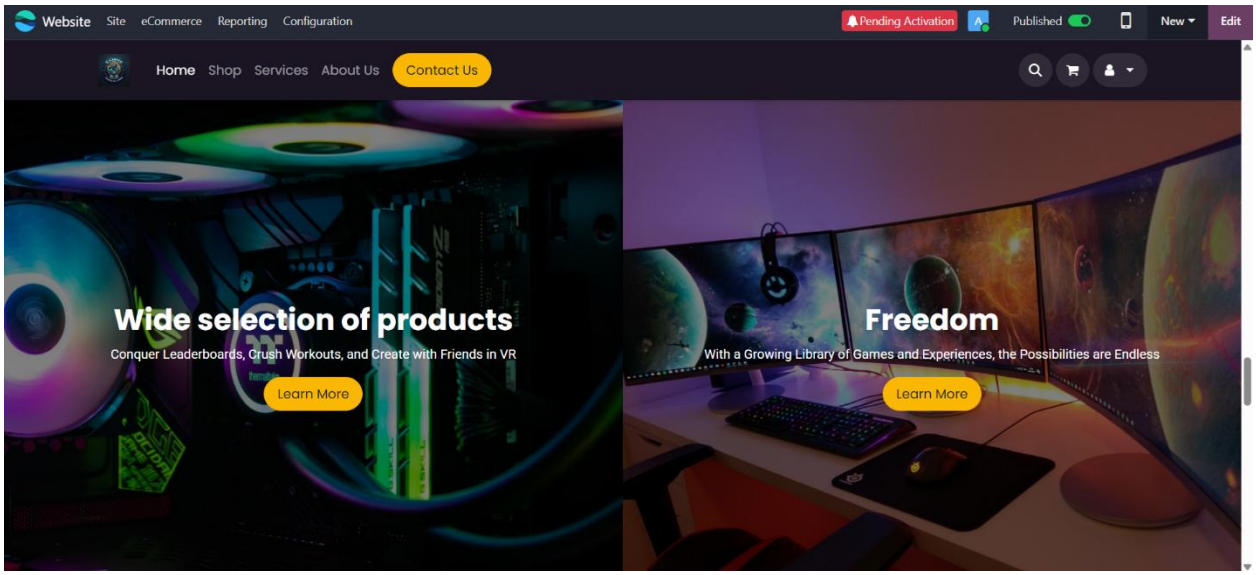

Website
Site
eCommerce
Reporting
Configuration
Pending Activation
Published
New
Edit
Home
Shop
Services
About Us
Contact Us
Low Price Guarantee
30 Days Online Returns
Standard Shipping
The Future of Computing
Discover more

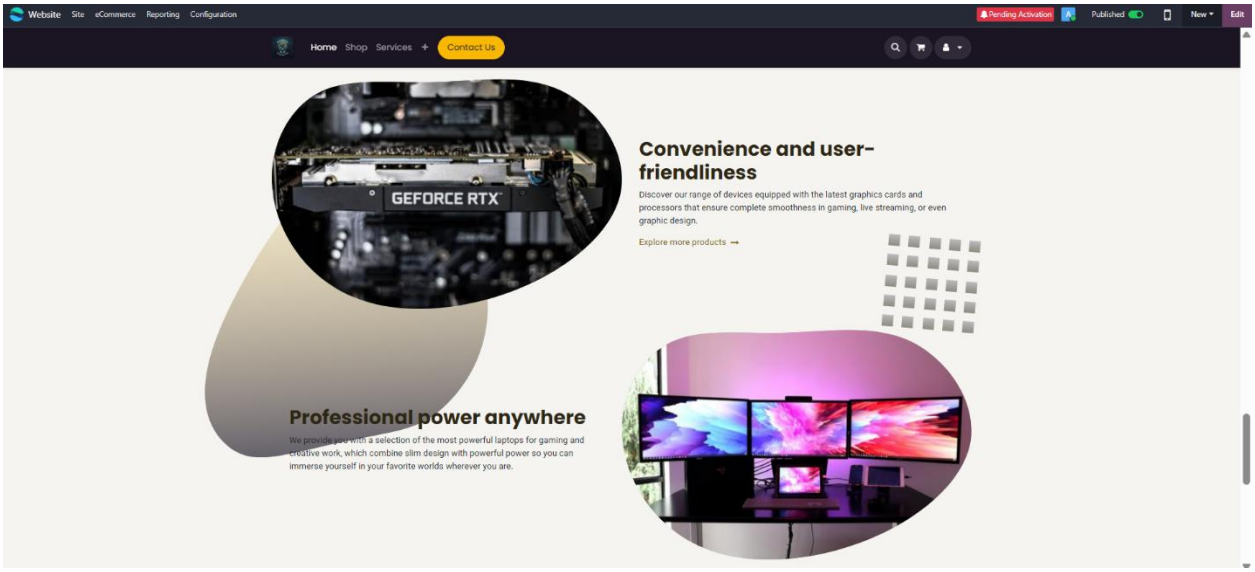

Website
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Pending Activation
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Shop
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About Us
Contact Us
Accessories
Components
Desktops
Gaming
Laptops
Monitors
Networking
Software
Unpublished

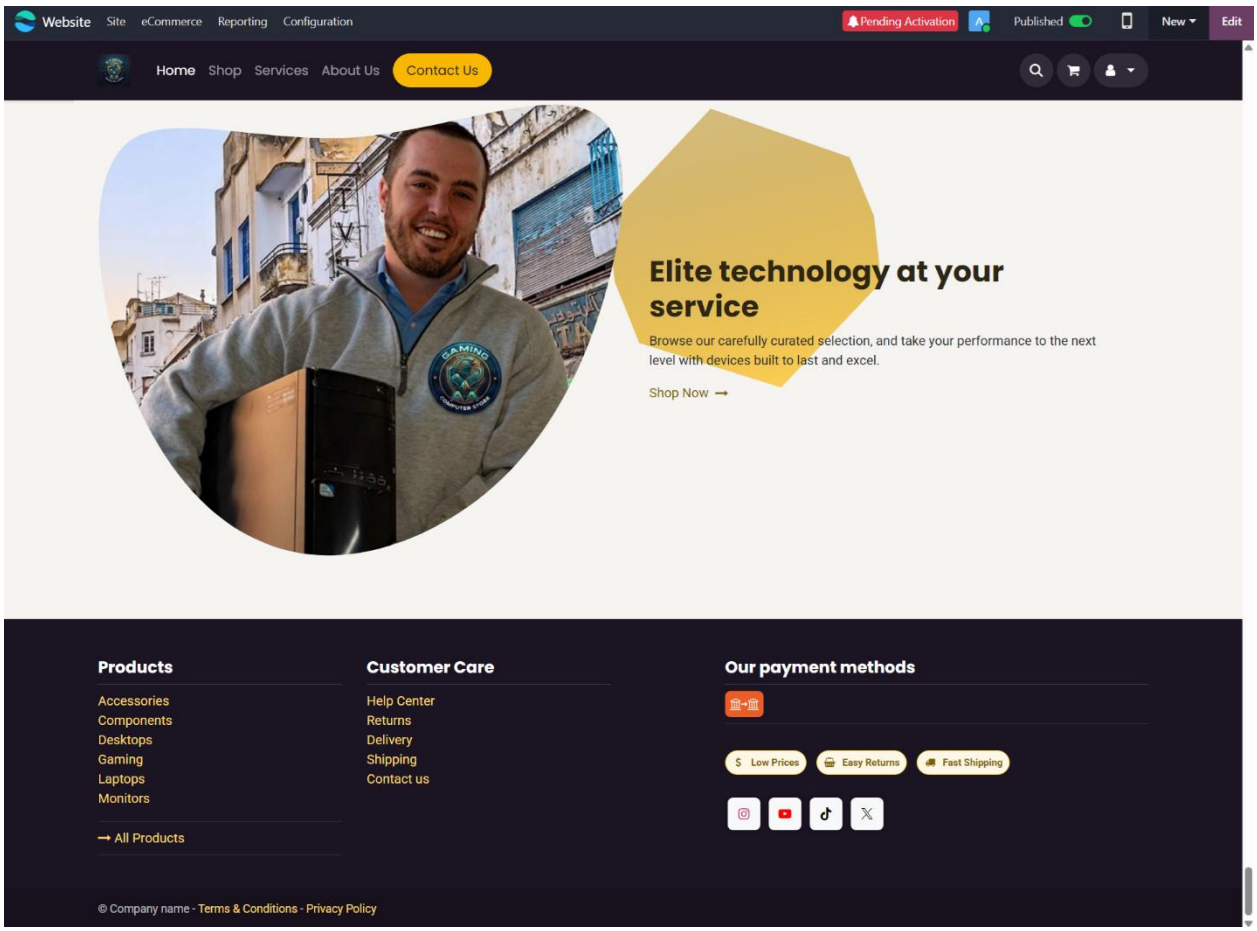

Website
Site
eCommerce
Reporting
Configuration
Pending Activation
Published
New
Edit
Home
Shop
Services
About Us
Contact Us
Level up with GamingAA
GamingAA is your ultimate destination for high-performance gaming gear. We specialize in curating the latest hardware and peripherals that push the boundaries of speed, graphics, and immersion, ensuring you stay ahead of the curve.
From custom-built rigs to professional-grade accessories, we provide cutting-edge technology solutions tailored to your specific gaming needs. Experience the next level of play with hardware that never compromises.
Learn more


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Configuration
Pending Activation
Published
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Edit
Home
Shop
Services
About Us
Contact Us
Experience the Future of Gaming
Step into an Extraordinary Gaming Universe with Exclusive Games


Website
Site
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Configuration
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Edit
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Shop
Services
About Us
Contact Us
Explore a new dimension
Step into the future of entertainment with GamingAA's premium selection of VR headsets and immersive tech.
Explore our curated collection of high-end accessories that prioritize both performance and comfort. Whether you're a professional gamer or looking for a new way to chill, we bring the next dimension of digital experiences right to your doorstep.
Learn more


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Site
eCommerce
Reporting
Configuration
Pending Activation
Published
New
Edit
Home
Shop
Services
About Us
Contact Us
Wide selection of products
Conquer Leaderboards, Crush Workouts, and Create with Friends in VR
Learn More
Freedom
With a Growing Library of Games and Experiences, the Possibilities are Endless
Learn More


Website
Site
eCommerce
Reporting
Configuration
Pending Activation
Published
New
Edit
Home
Shop
Services
Contact Us
GEFORCE RTX
Convenience and user-friendliness
Discover our range of devices equipped with the latest graphics cards and processors that ensure complete smoothness in gaming, live streaming, or even graphic design.
Explore more products
Professional power anywhere
We provide you with a selection of the most powerful laptops for gaming and creative work, which combine slim design with powerful power so you can immerse yourself in your favorite worlds wherever you are.


Website
Site
eCommerce
Reporting
Configuration
Pending Activation
Published
New
Edit
Home
Shop
Services
About Us
Contact Us
Elite technology at your service
Browse our carefully curated selection, and take your performance to the next level with devices built to last and excel.
Shop Now
Products
Accessories
Components
Desktops
Gaming
Laptops
Monitors
All Products
Customer Care
Help Center
Returns
Delivery
Shipping
Contact us
Our payment methods
Low Prices
Easy Returns
Fast Shipping
© Company name - Terms & Conditions - Privacy Policy

Part II. Mini-Project TP ERP

I. Project Introduction

This project aims to introduce and practically implement an Enterprise Resource Planning (ERP) system using the Odoo platform. This report focuses on creating a virtual company and managing its core operations, such as sales, purchasing, inventory, and accounting.

II. Introducing Intoy

Intoy is an online platform specializing in the sale of toys for children of all ages. It offers a wide range of products, including dolls, teddy bears, toy vehicles, as well as educational and interactive games. Intoy's primary objective is to provide quality products that meet children's entertainment and developmental needs.

In addition to selling products, Intoy can offer complementary services such as gift wrapping to enhance the customer experience. The management of these activities relies on an ERP system that efficiently organizes the processes of purchasing from suppliers, inventory management in warehouses, and sales and invoicing processes for customers.

Thus, Intoy represents a concrete example of an integrated commercial business, where purchasing, stock management, and sales flows are interconnected to ensure efficient management and greater customer satisfaction.

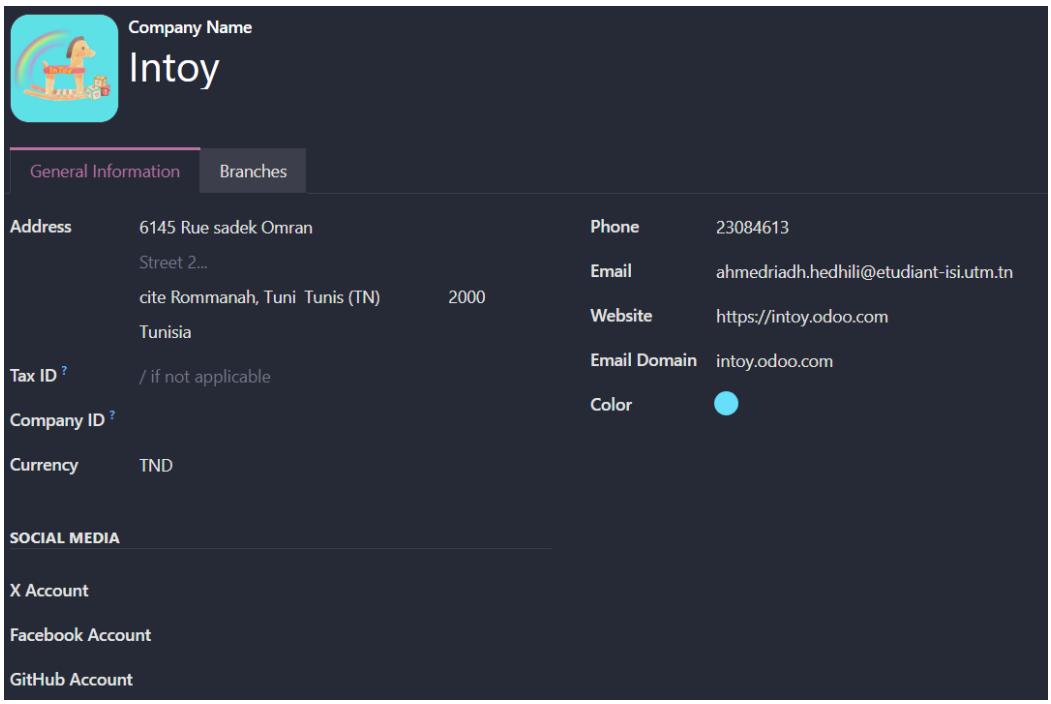
III. Activity 1: Company setup

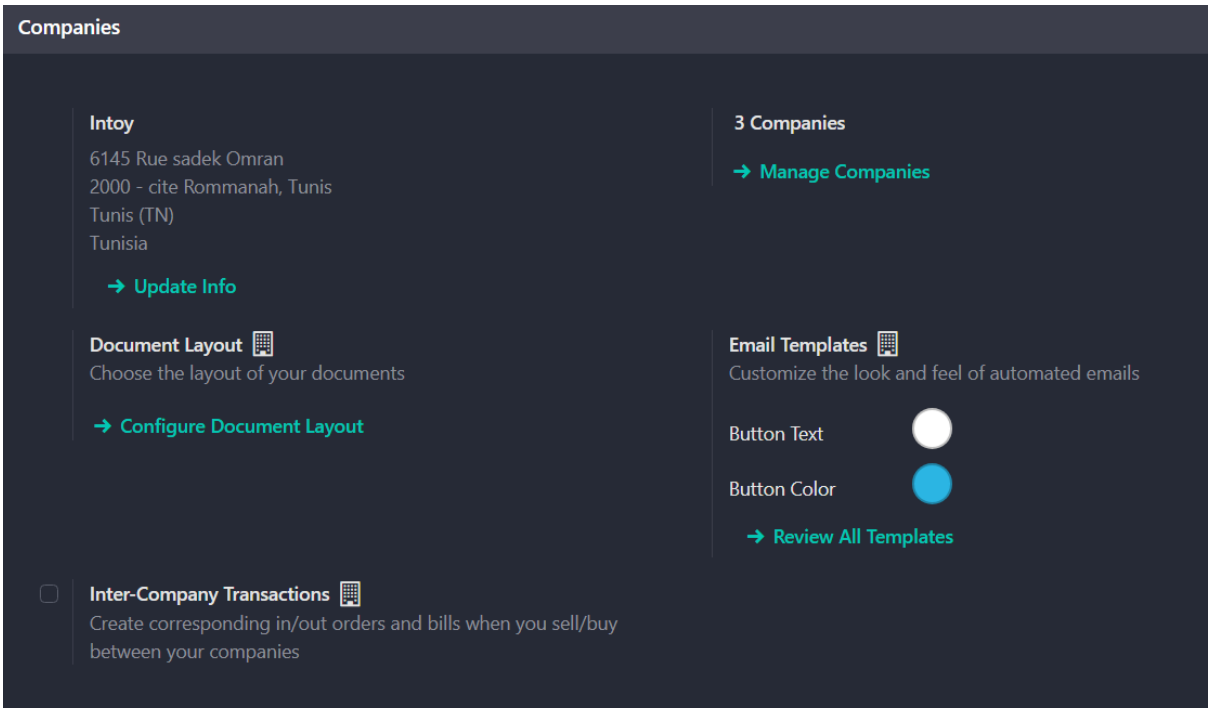
Basic company information:

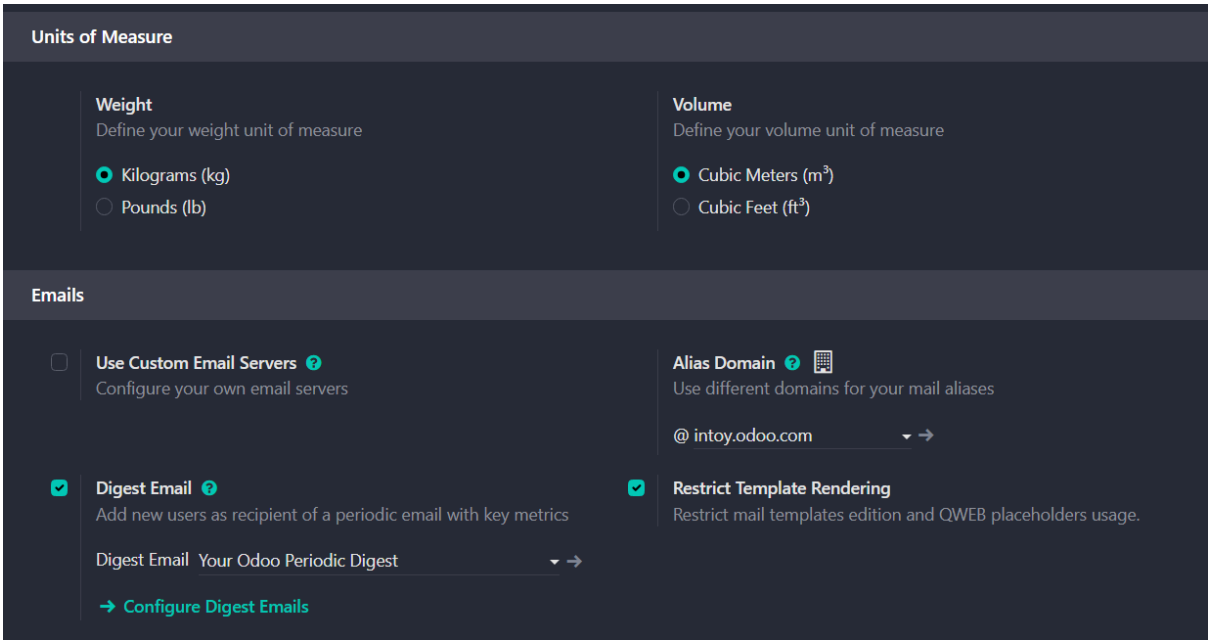
https://intoy.odoo.com

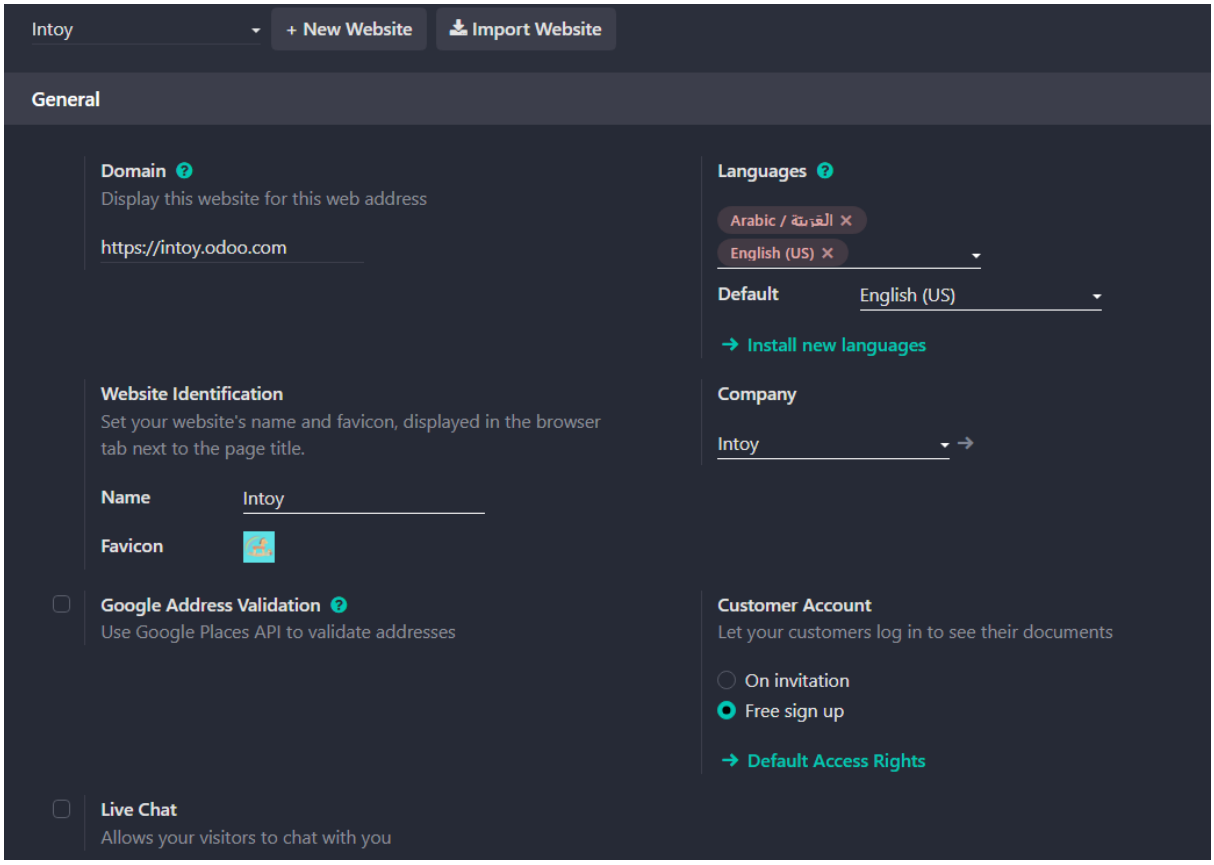
- Company name: Intoy
- City: Tunis
- Country: Tunisia
- Currency: TND (Tunisian Dinar)
- Phone: +216 23 084 613
- Email: info@ intoy.odoo.com
- Text slogan: “Gifts of joy for every child”

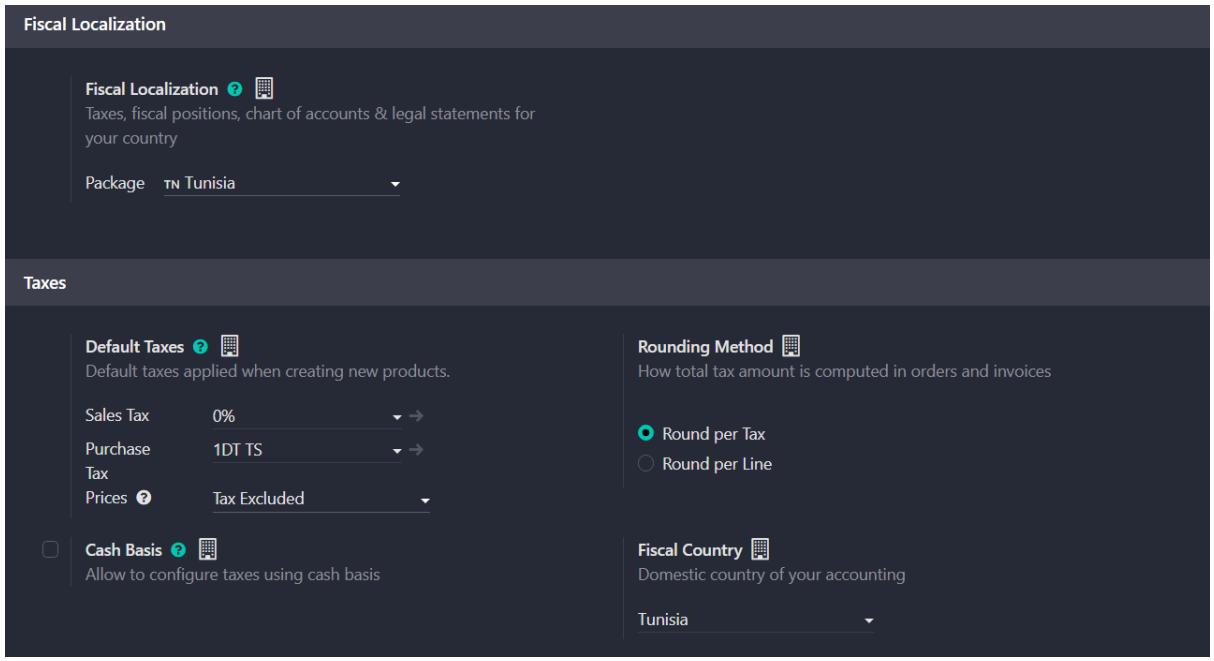
The data has been successfully saved in the system.




Companies
Intoy
6145 Rue sadek Omran
2000 - cite Rommanah, Tunis
Tunis (TN)
Tunisia
→ Update Info
3 Companies
→ Manage Companies
Document Layout
Choose the layout of your documents
→ Configure Document Layout
Email Templates
Customize the look and feel of automated emails
Button Text
Button Color
→ Review All Templates
Inter-Company Transactions
Create corresponding in/out orders and bills when you sell/buy between your companies


Units of Measure
Weight
Define your weight unit of measure
Kilograms (kg)
Pounds (lb)
Volume
Define your volume unit of measure
Cubic Meters (m³)
Cubic Feet (ft³)
Emails
Use Custom Email Servers
Configure your own email servers
Alias Domain
Use different domains for your mail aliases
@ intoy.odoo.com
Digest Email
Add new users as recipient of a periodic email with key metrics
Digest Email Your Odoo Periodic Digest
→ Configure Digest Emails
Restrict Template Rendering
Restrict mail templates edition and QWEB placeholders usage.

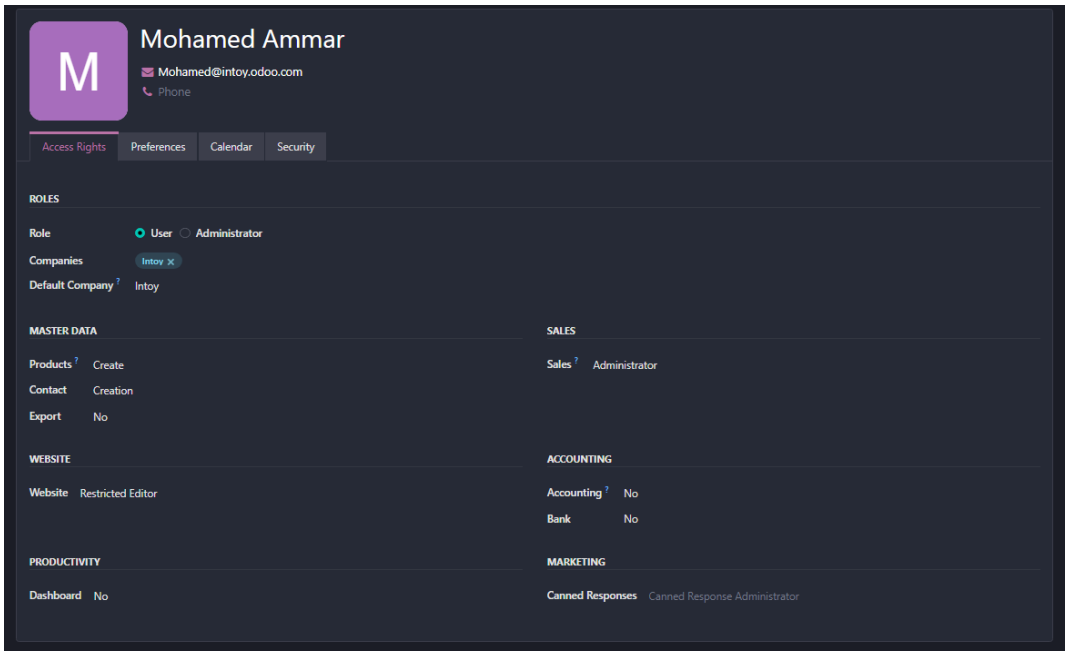

Intoy
+ New Website
Import Website
General
Domain
Display this website for this web address
https://intoy.odoo.com
Languages
Arabic / العربية
English (US)
Default
English (US)
Install new languages
Website Identification
Set your website's name and favicon, displayed in the browser tab next to the page title.
Name
Intoy
Favicon
Company
Intoy
Google Address Validation
Use Google Places API to validate addresses
Customer Account
Let your customers log in to see their documents
On invitation
Free sign up
Default Access Rights
Live Chat
Allows your visitors to chat with you


Fiscal Localization
Fiscal Localization
Taxes, fiscal positions, chart of accounts & legal statements for your country
Package
TN Tunisia
Taxes
Default Taxes
Default taxes applied when creating new products.
Sales Tax
0%
Purchase Tax
1DT TS
Prices
Tax Excluded
Rounding Method
How total tax amount is computed in orders and invoices
Round per Tax
Round per Line
Cash Basis
Allow to configure taxes using cash basis
Fiscal Country
Domestic country of your accounting
Tunisia

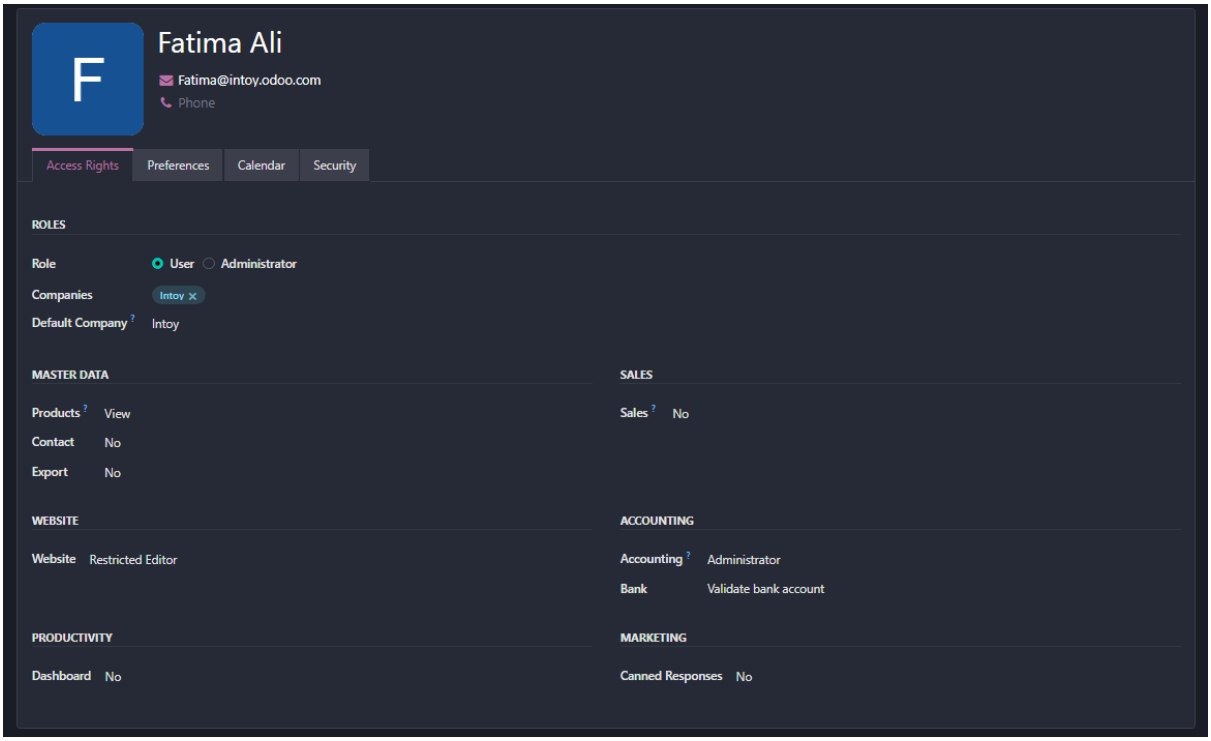
IV. Activity 2: User Management

Create different users with different permissions to work in the store. Each employee will have a specific role and certain powers.

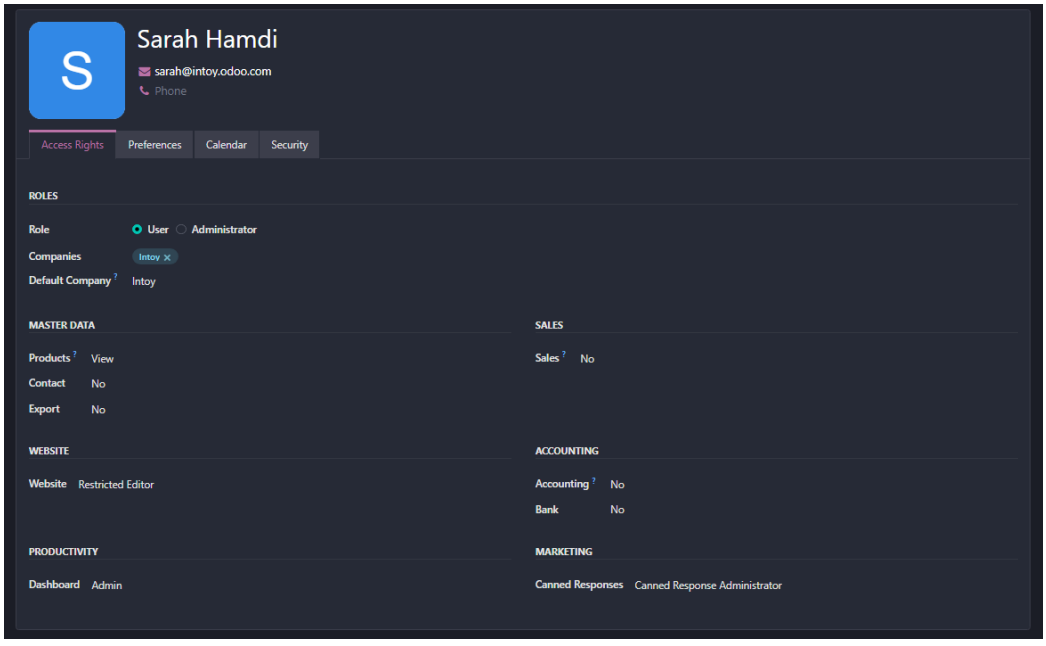
- **First user:**
  - Name: Mohamed Ammar
  - Email: Mohamed@intoy.odoo.com
  - Username: Med_sales
  - Role: Sales Manager
  - Powers: He can create sales orders and manage customers

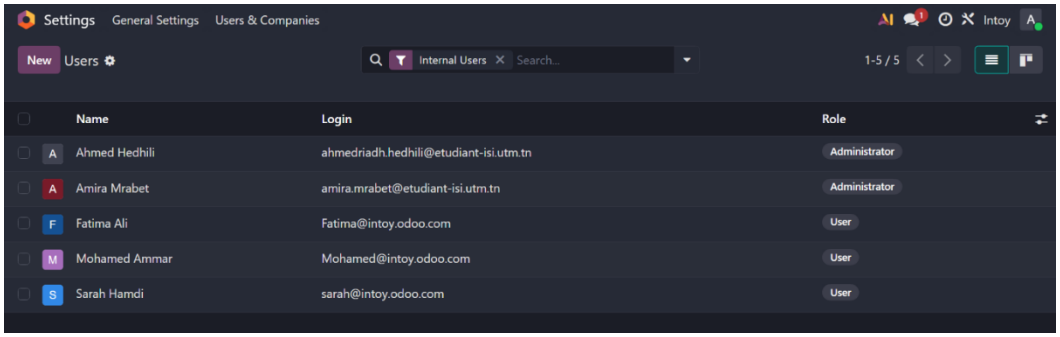


- **Second user:**
  - Name: Fatima Ali
  - Email: Fatima@intoy.odoo.com
  - Username: fatim_accountant
  - Role: Accountant
  - Powers: She can manage invoices and payments



- **Third user:**
  - Name: Sarah Hamdi
  - Email: sarah@intoy.odoo.com
  - Username: sar_marketing
  - Role: Marketing
  - Powers: She can manage promoting products to the right audience





V. Activity 3: Module Installation

Install the essential modules needed to manage the toy and doll shop. Each module adds new features to the system.

The following four core modules must be installed:

1. CRM (Customer Relationship Management)

- Uses: Managing customer information and communicating with customers
- Features: Customer tracking, sales opportunities, communication

2. Sales

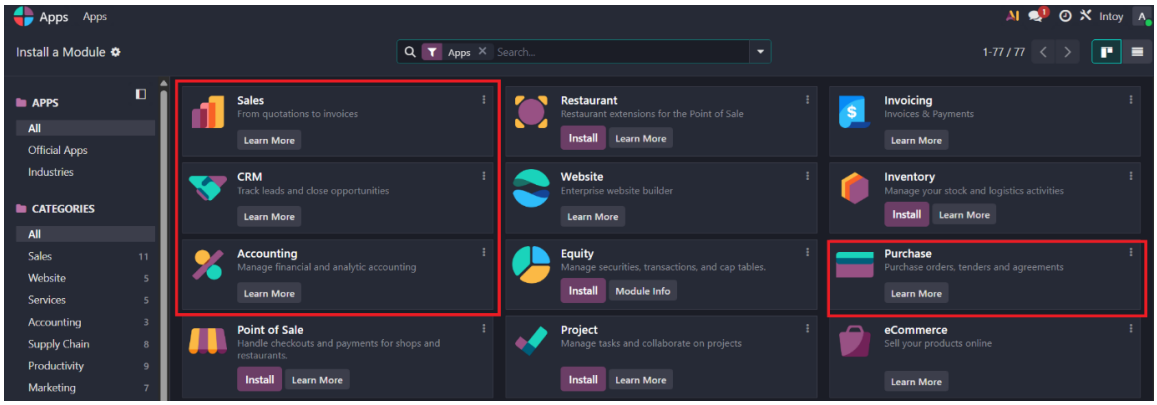
- Uses: Creating sales orders and product deliveries
- Features: Sales orders, invoices, deliveries

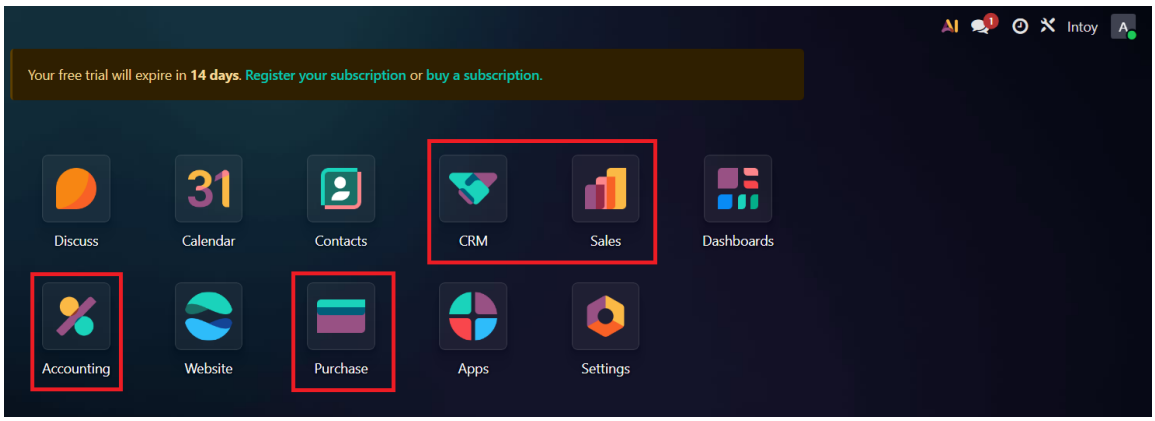
3. Purchases

- Uses: Creating purchase orders from suppliers
- Features: Purchase orders, receiving goods, supplier invoices

4. Accounting

- Uses: Managing invoices, payments, and financial reports
- Features: Accounting entries, reports, account management

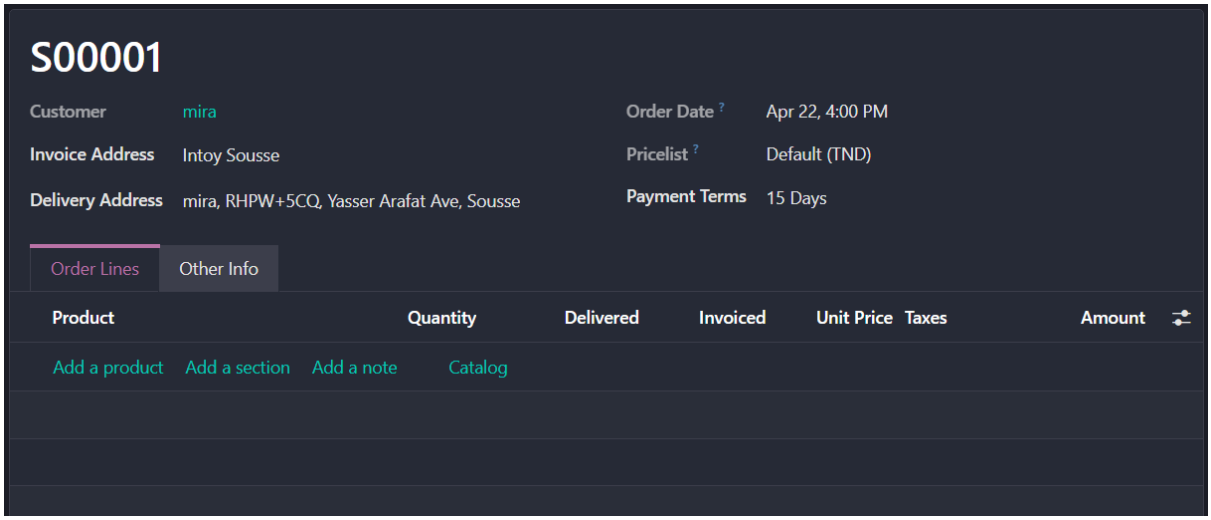


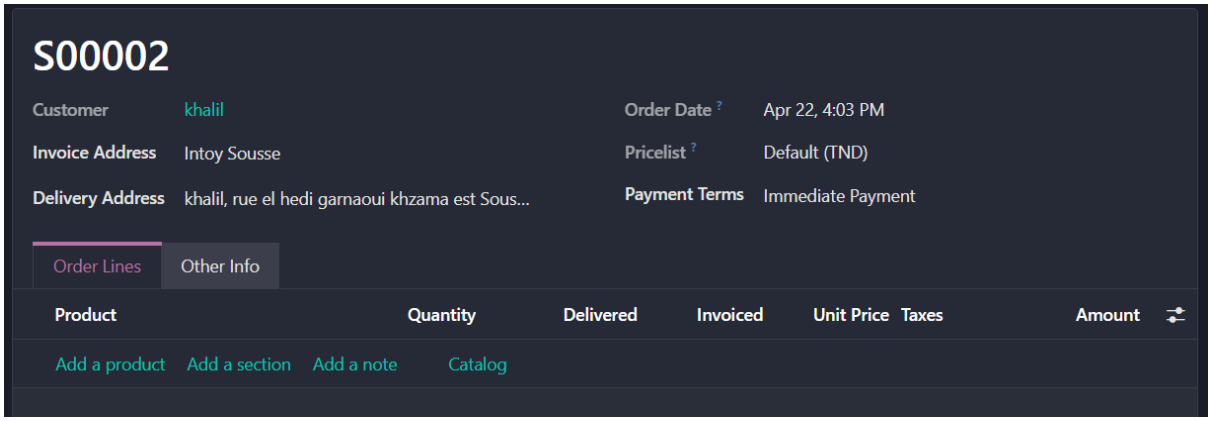


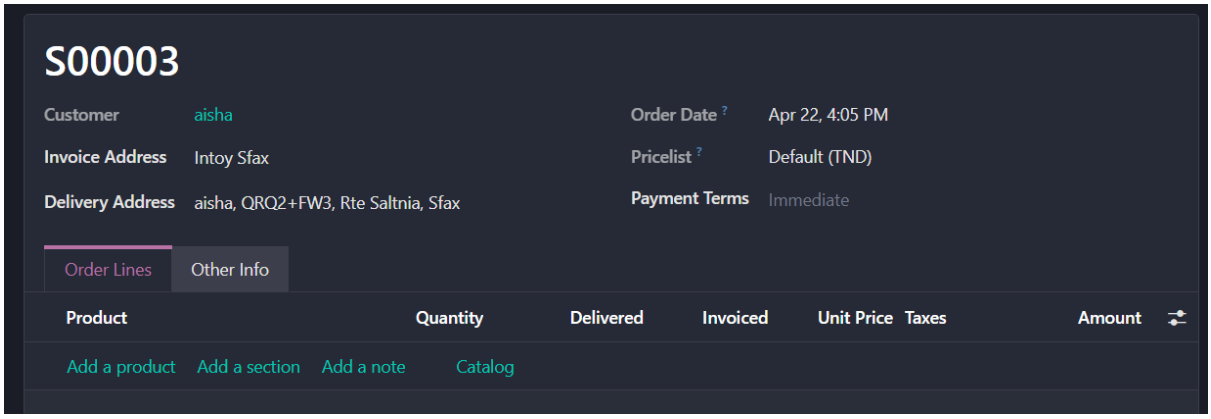
VI. Activity 4: Partner Management

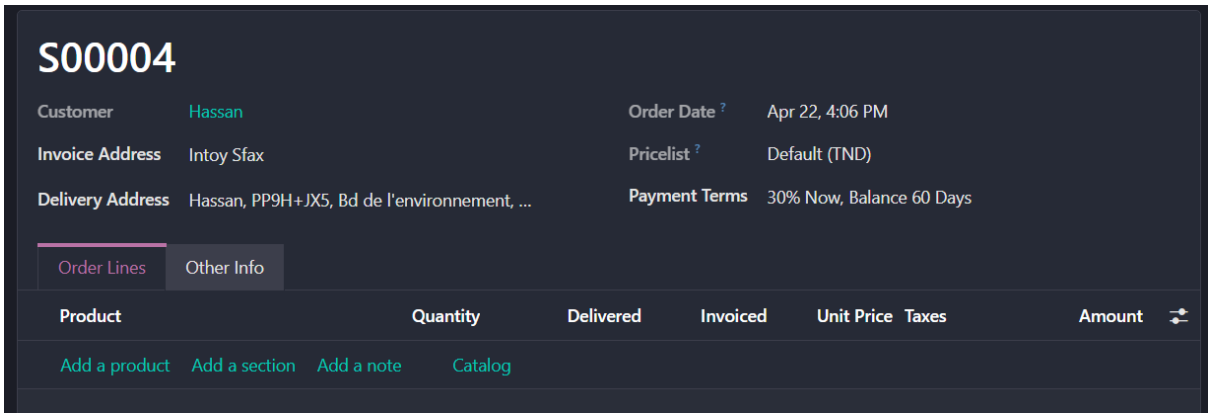
To define partners in Odoo, the process typically involves navigating through the respective modules:

- For Customers: Sales > Customers > Create

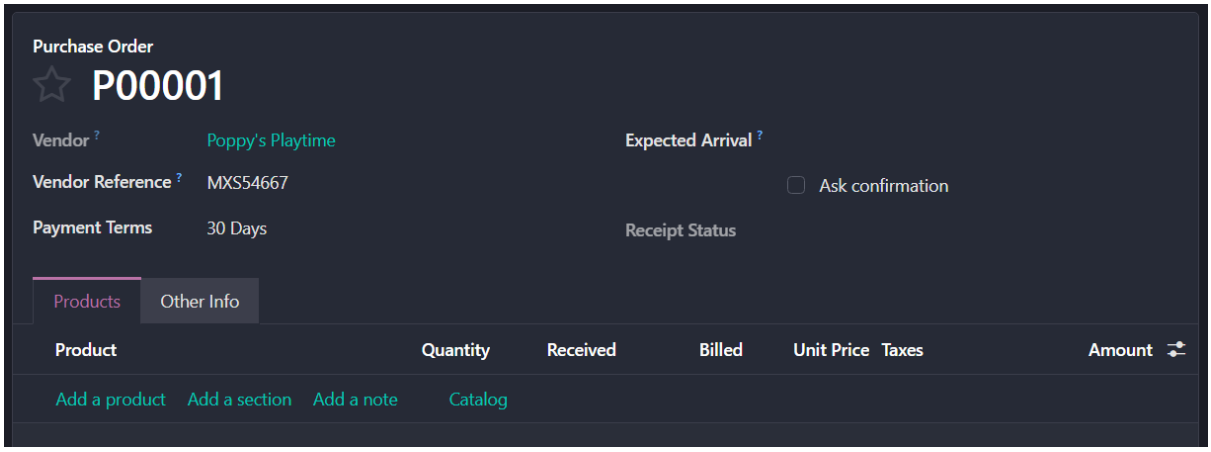


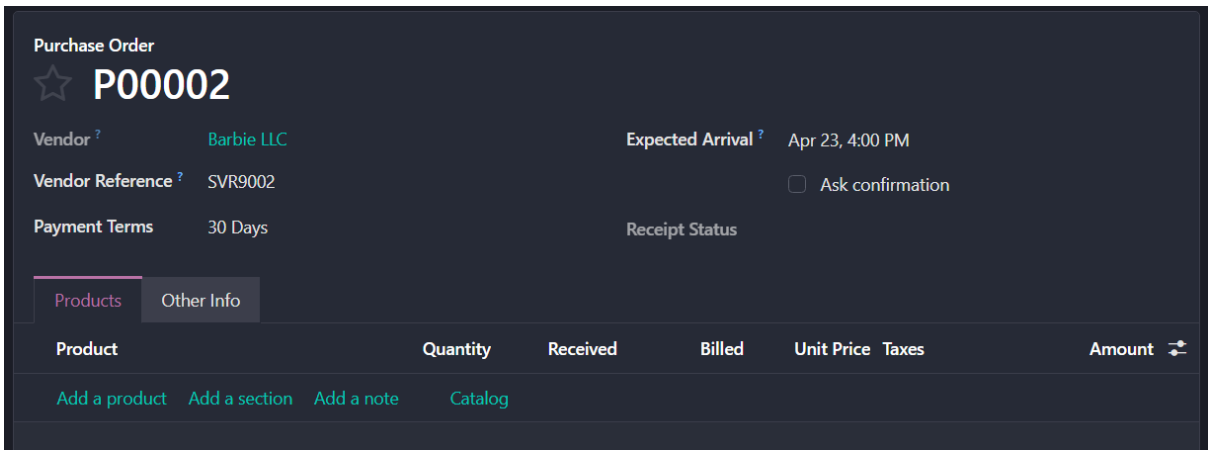


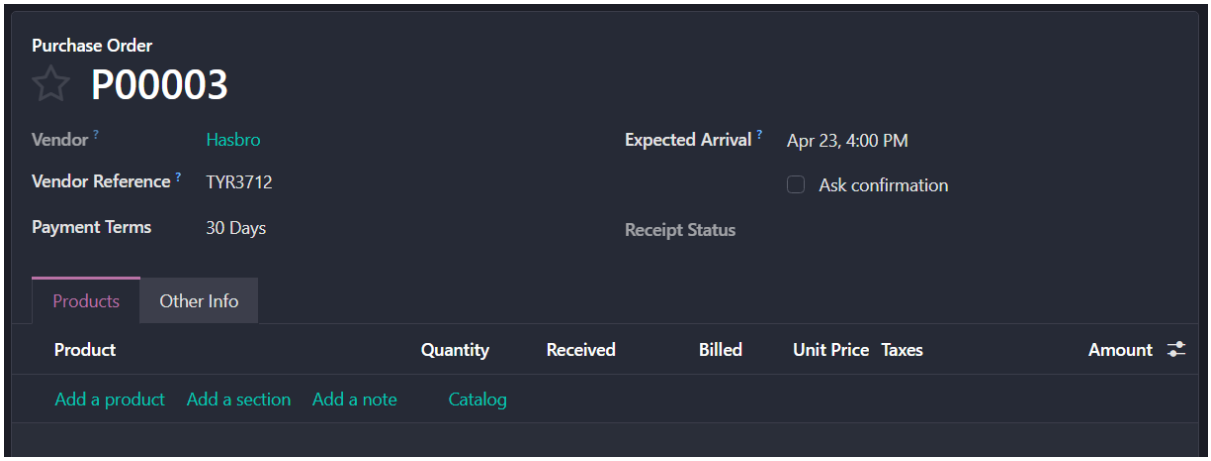


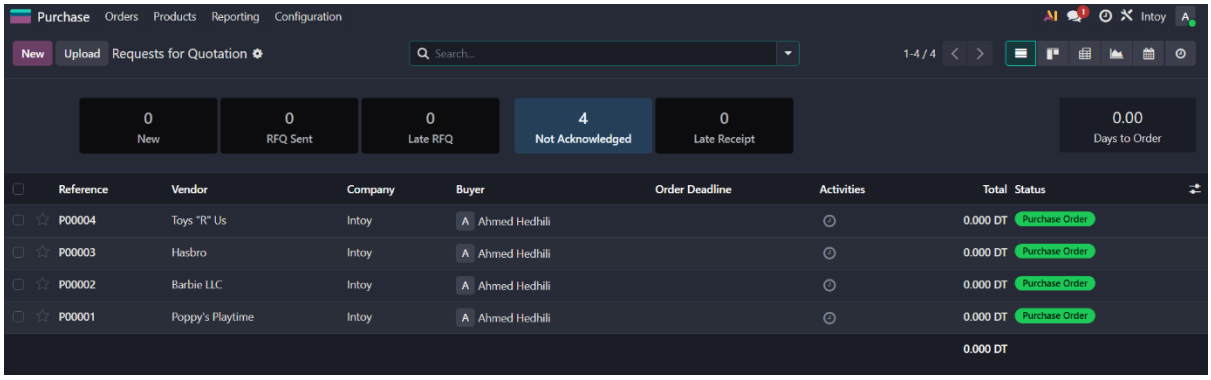


❖ For Suppliers: Purchases > Suppliers > Create





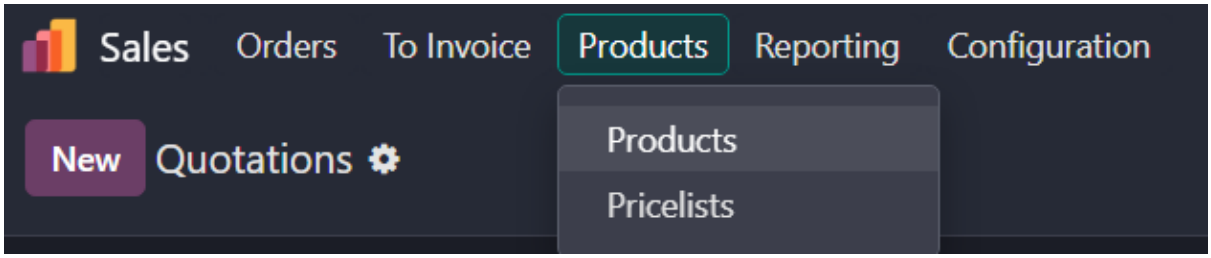


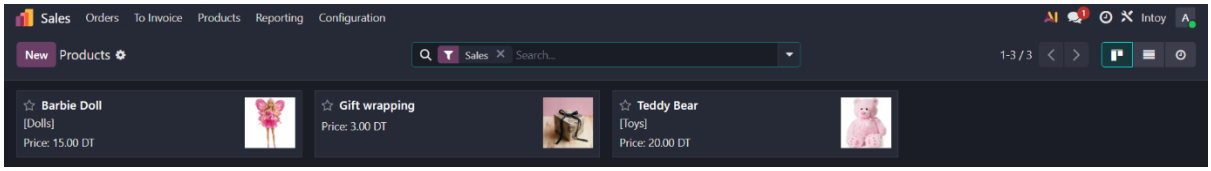


VII. Activity 5: Product Management

To add products and services in Odoo, users typically navigate through the following menu paths:

➢ For Sales: Sales > Products > Create





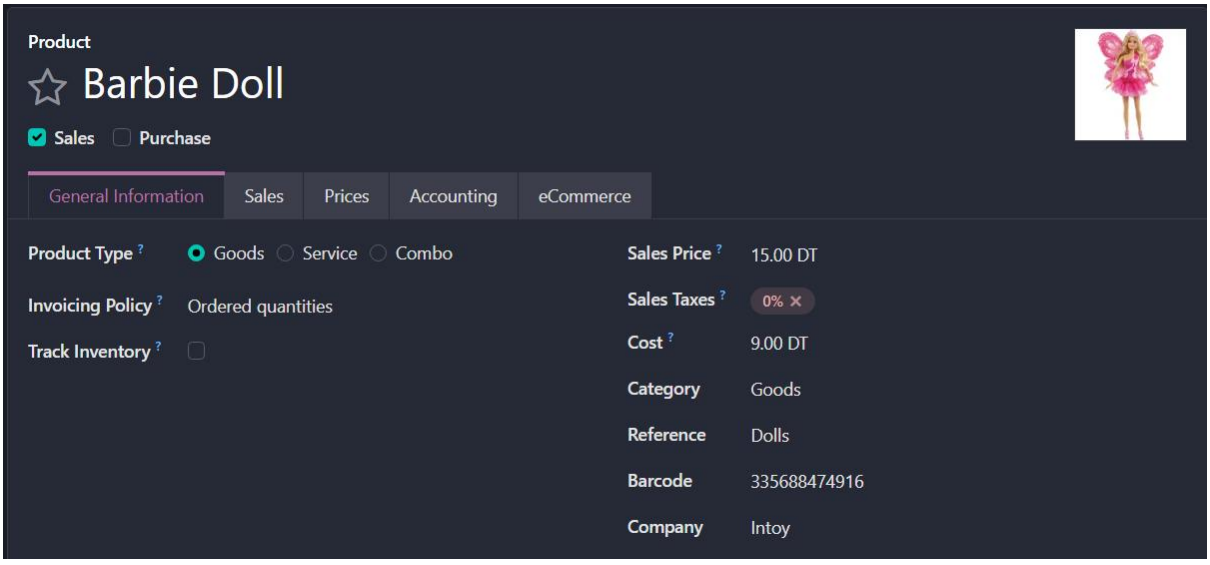
1- Product Name:

Barbie Doll (Pink Dress with Butterfly Wings).

- **Saleable/Purchasable Status:** Saleable (a physical item that can be stored in inventory).
- **Product Type:** Storable product.
- **Sales Price:** 15 TND.
- **Unit of Measure:** Unit (piece).
- **Supplier Information:** Supplier Name: Barbie LLC.

  Purchase Price: 9 TND.

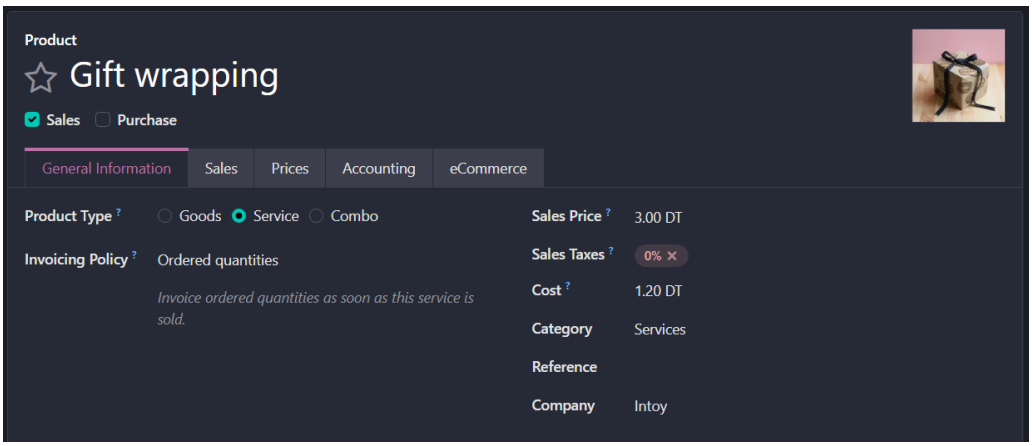
  Delivery Lead Time: 5–10 days.



2- Service Name:

Gift wrapping.

- **Saleable/Purchasable Status:** Saleable.
- **Product Type:** Service.
- **Sales Price:** 3 TND per item.
- **Unit of Measure:** Unit (Service per item).
- **Supplier Information:** Not required.



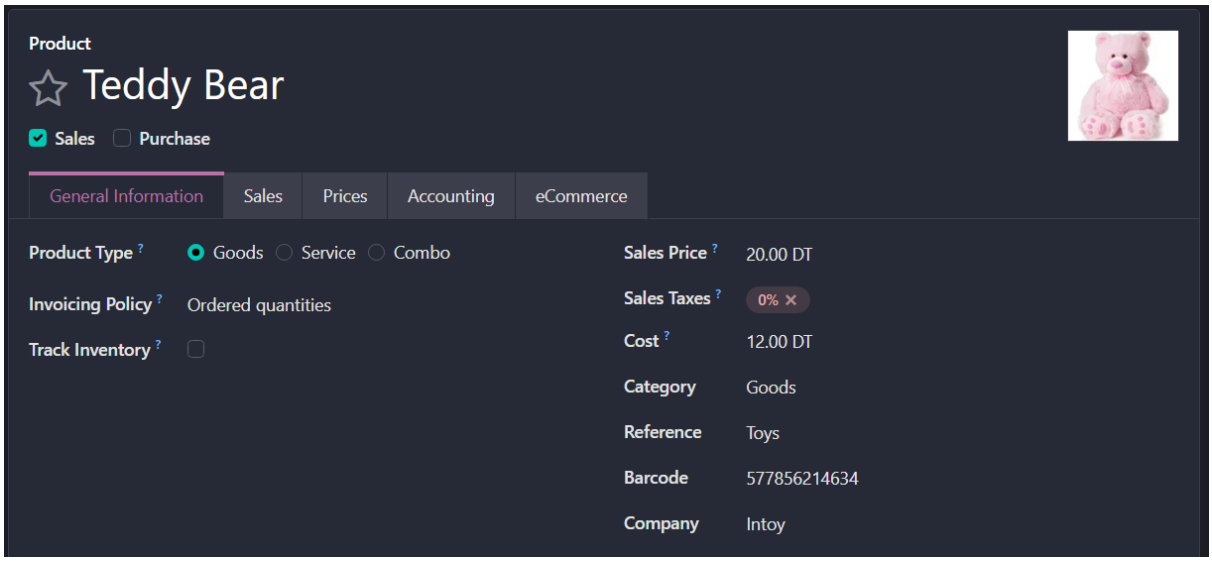
3- Product Name:

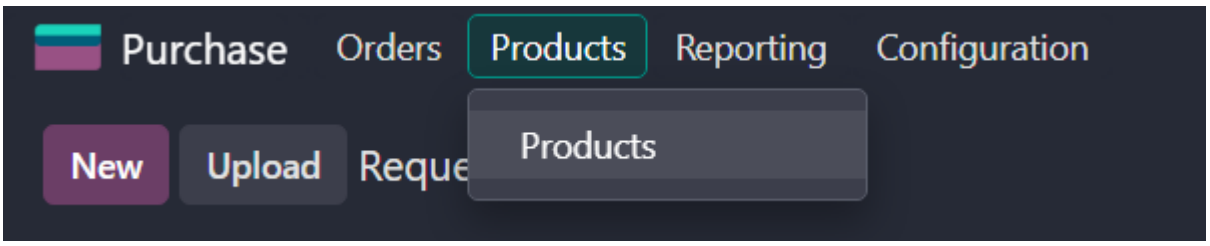
Small Plush Teddy Bear (Fluffy Faux Fur with Ribbon Bow) .

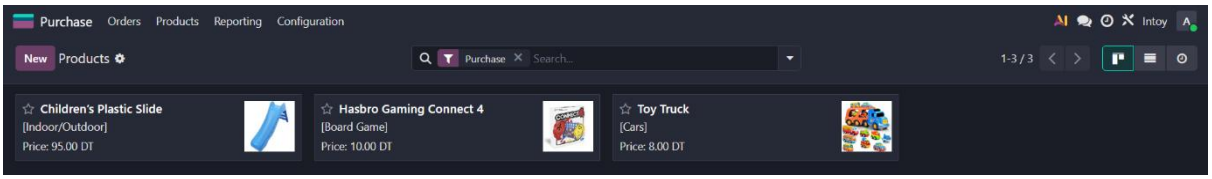
- **Saleable/Purchasable Status:** Saleable.
- **Product Type:** Storable product.
- **Sales Price:** 20 TND.
- **Unit of Measure:** Unit (piece).
- **Supplier Information:** Supplier Name: Toys "R" Us.

  Purchase Price: 12 TND.

  Delivery Lead Time: 5–10 days.







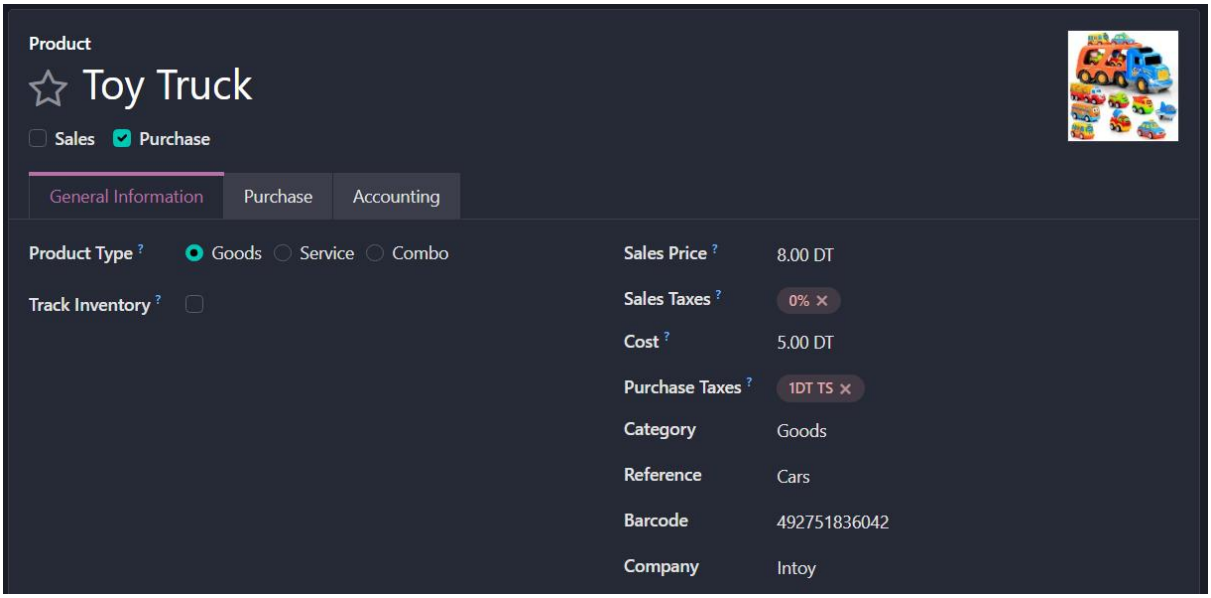
1- Product Name:

Toy Transport Truck with Mini Vehicles set.

- **Saleable/Purchasable Status:** Purchasable.
- **Product Type:** Storable product.
- **Sales Price:** 8 TND.
- **Unit of Measure:** Unit (Set).
- **Supplier Information:** Supplier Name: Toys "R" Us.

  Purchase Price: 5 TND.

  Delivery Lead Time: 5–10 days.



2- Product Name:

Children's Foldable Indoor/Outdoor Plastic Slide (Light Blue).

- **Saleable/Purchasable Status:** Purchasable.
- **Product Type:** Storable product.
- **Sales Price:** 95 TND.
- **Unit of Measure:** Unit (Set).
- **Supplier Information:** Supplier Name: Toys "R" Us.

  Purchase Price: 57 TND.

  Delivery Lead Time: 10 days



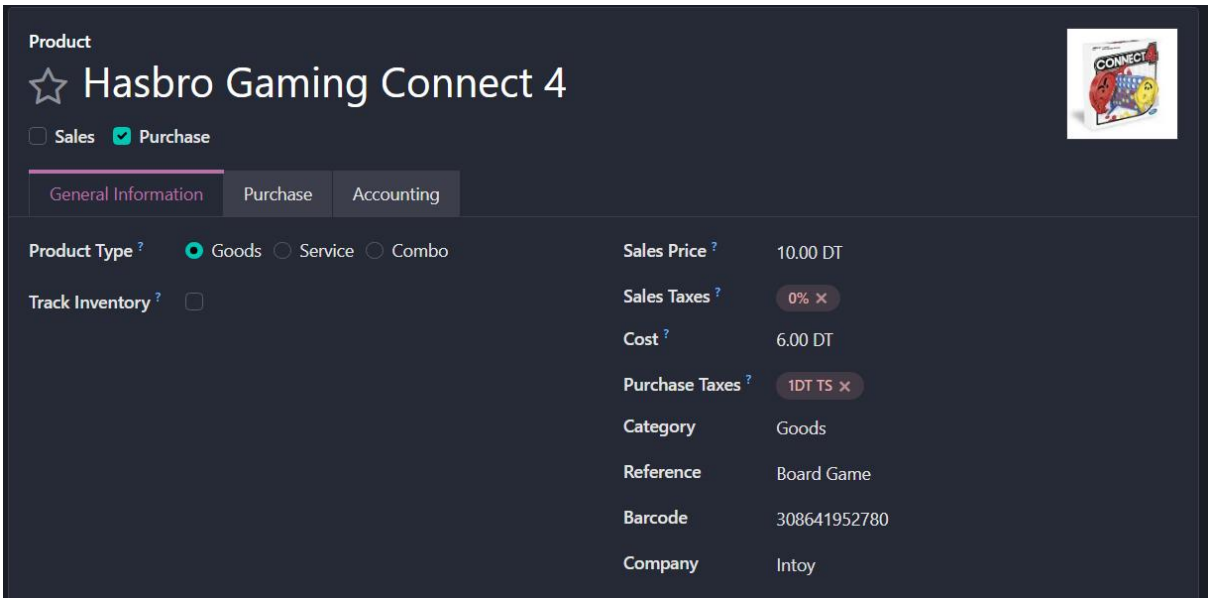
3- Product Name:

Hasbro Gaming Connect 4 Classic Strategy Board Game

- **Saleable/Purchasable Status:** Purchasable.
- **Product Type:** Storable product.
- **Sales Price:** 10 TND.
- **Unit of Measure:** Unit (as a single complete box set).
- **Supplier Information:** Supplier Name: Hasbro.

  Purchase Price: 6 TND.

  Delivery Lead Time: 5–10 days

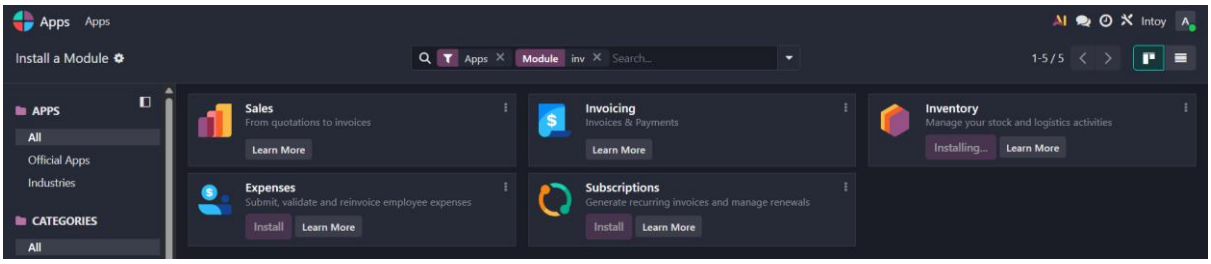


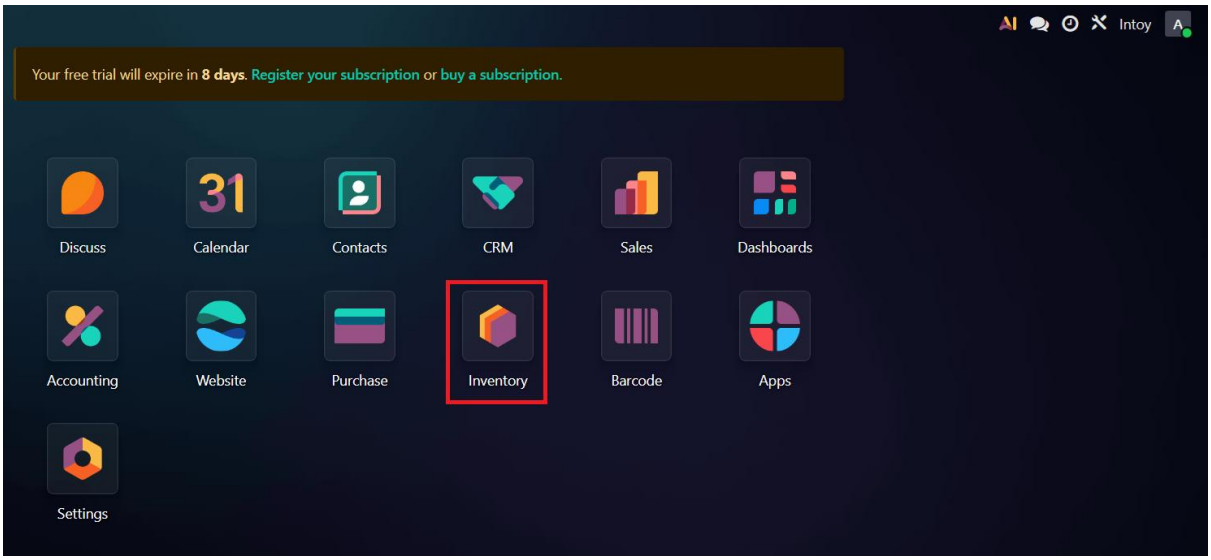
VIII. Activity 6: Inventory & Warehouse Management

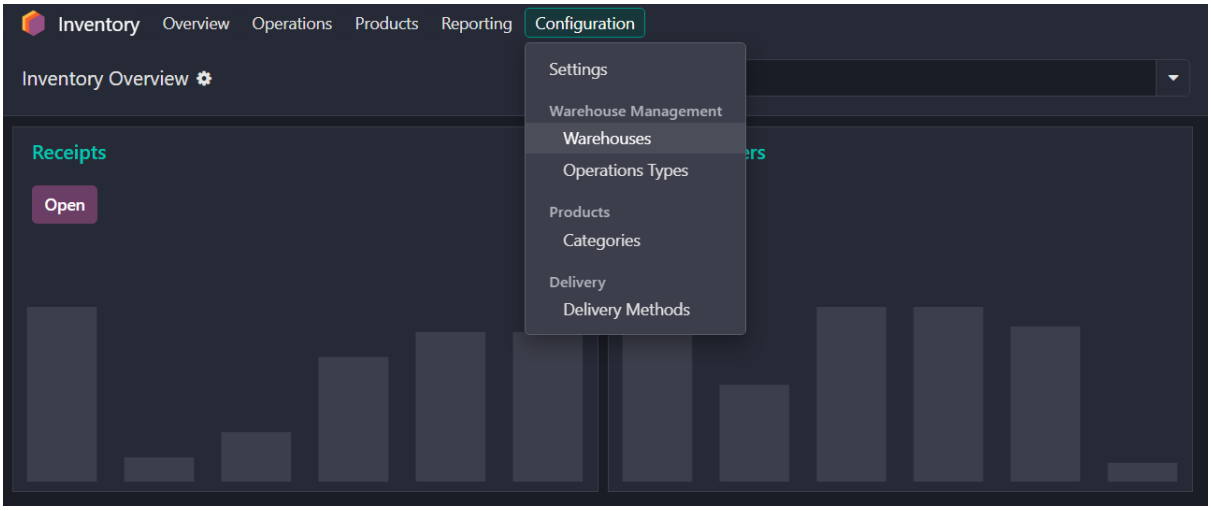
To manage the toys and dolls efficiently, we need to define specific storage locations. In Odoo, this is done via:

Inventory > Configuration > Warehouses.

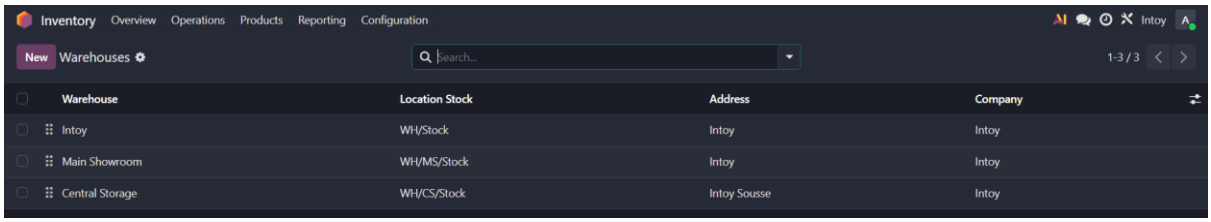
First, we need to install the Inventory's ERP model from Apps:



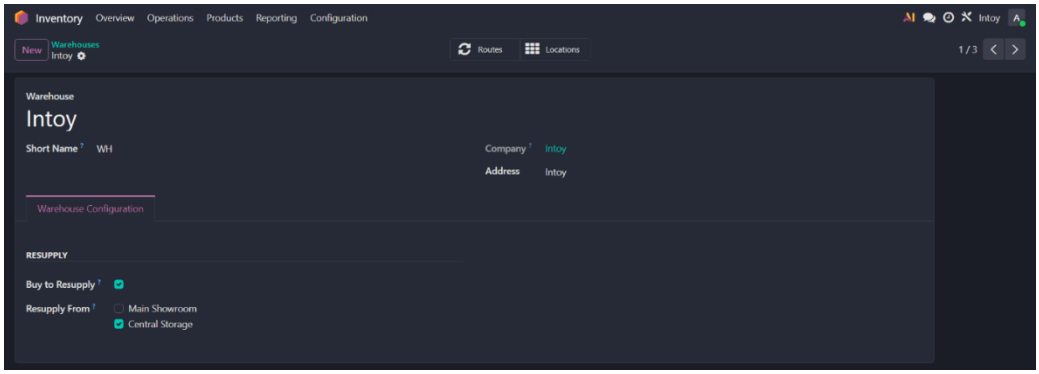




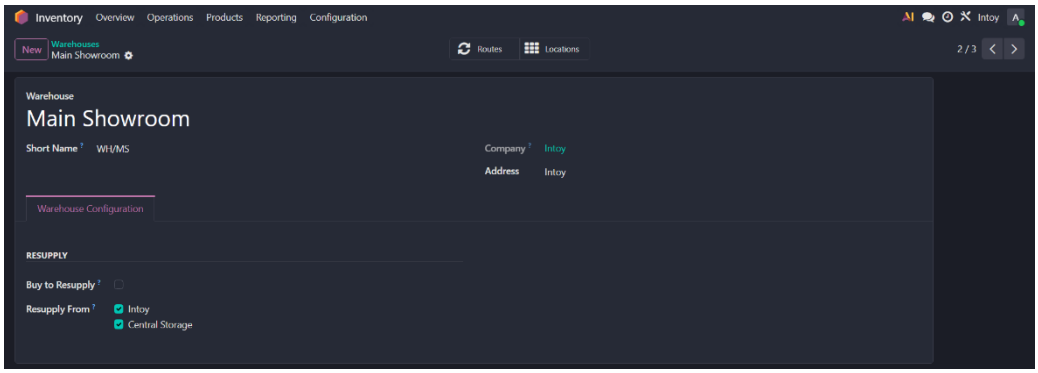
Proposed Warehouses:



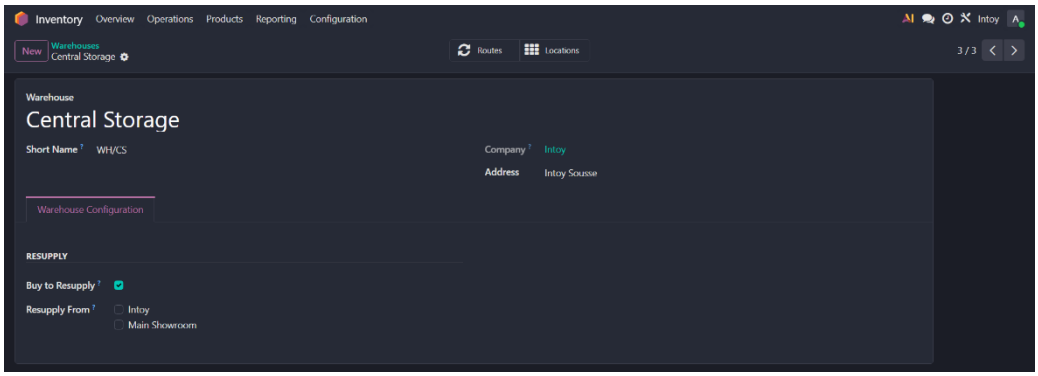
1. **Intoy Warehouse:** the main and virtual warehouse for our store; it receives, stores and ships toys and dolls, and manages all of our company's inventory movements.



2. **Main Showroom (WH/Stock):** The primary display area where customers browse and buy toys directly.



3. **Central Storage (WH/Storage):** A back-end warehouse used for storing bulk quantities of dolls and large toy sets before they are moved to the showroom.

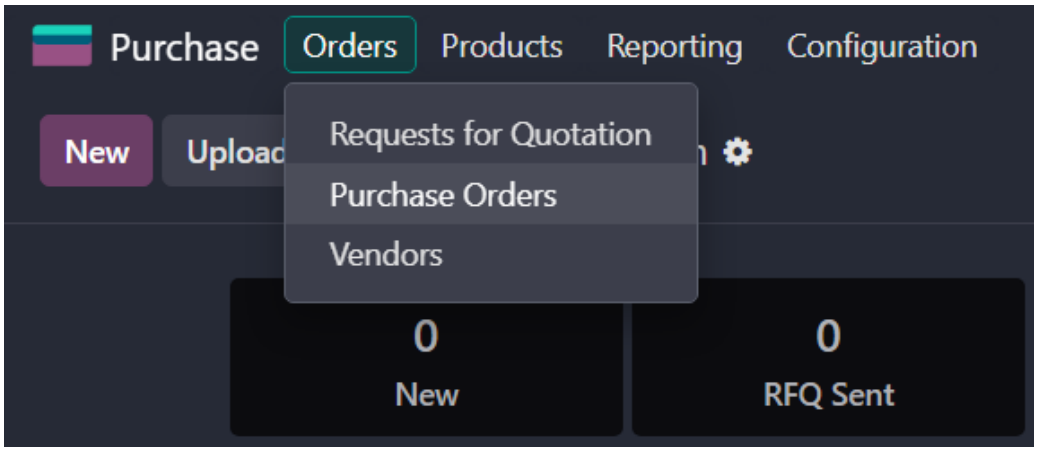


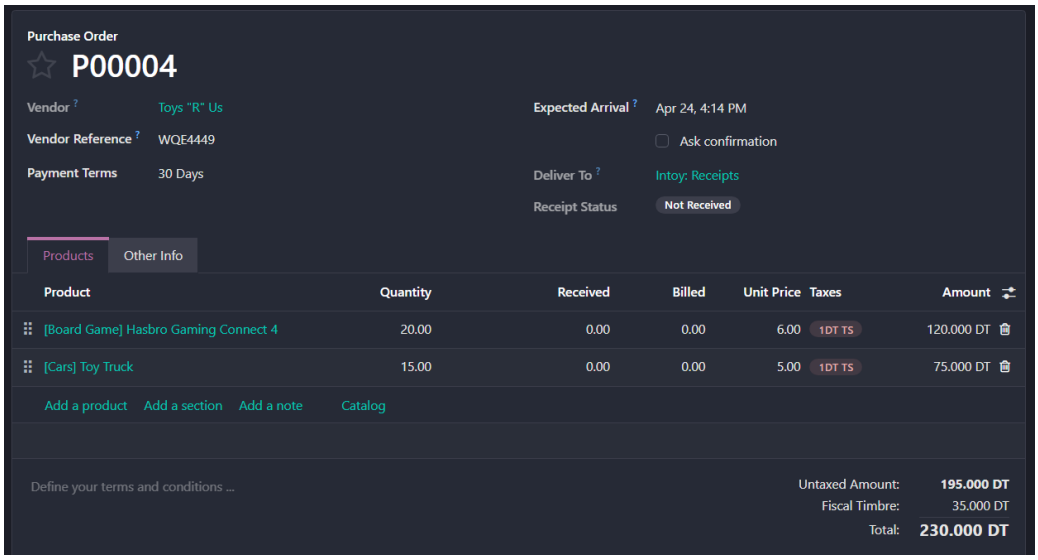
IX. Activity 7: Purchasing Workflow

In this phase, we simulate the process of replenishing our toy stock from suppliers.

1. Create Purchase Order (PO):

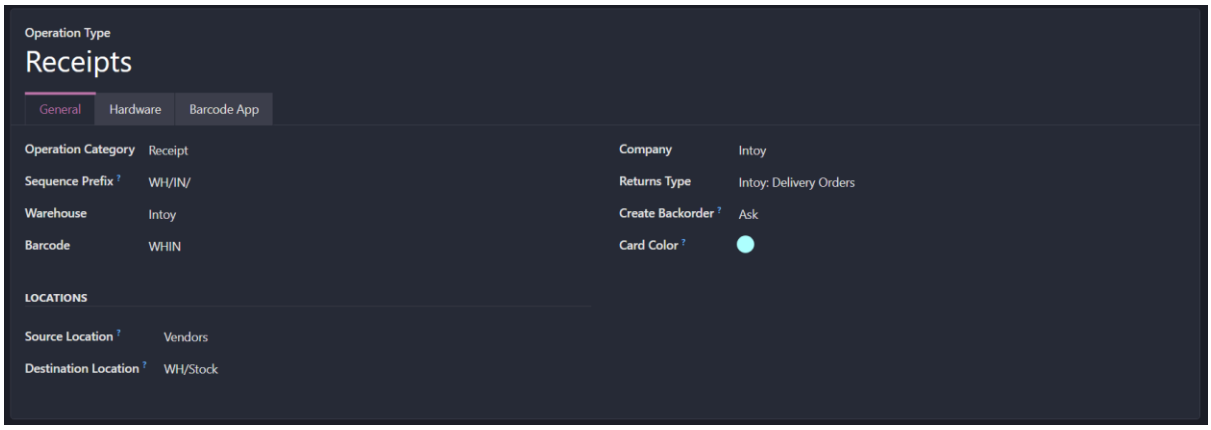
Purchases > Orders > Purchase Orders > Create





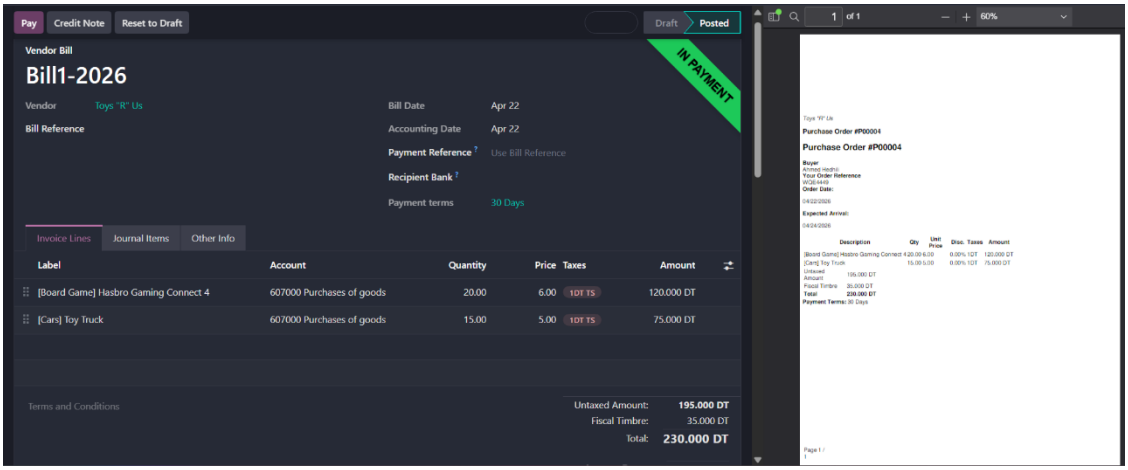
2. Receive Products:

- Once confirmed, click the **Receive Products** button (this adds the toys to your **Intoy** warehouse).
- Click **Validate** to confirm you received the physical goods.

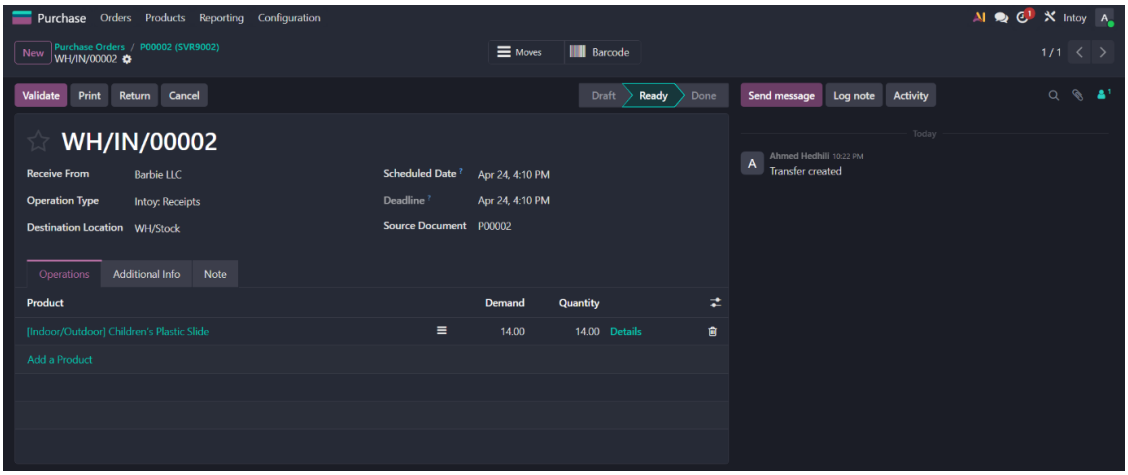


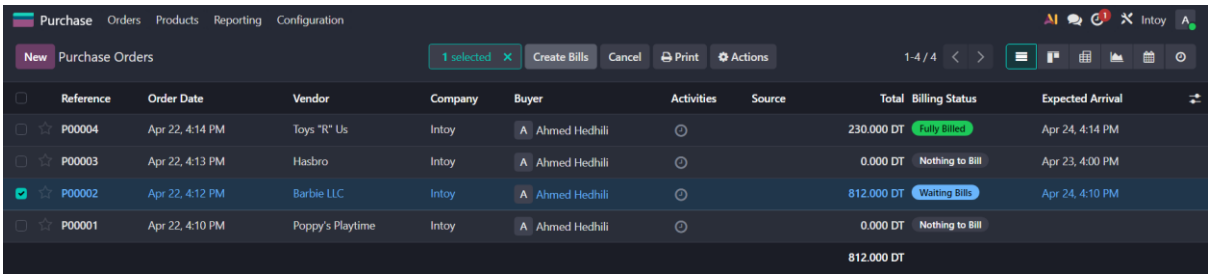
3. Vendor Bill & Payment:

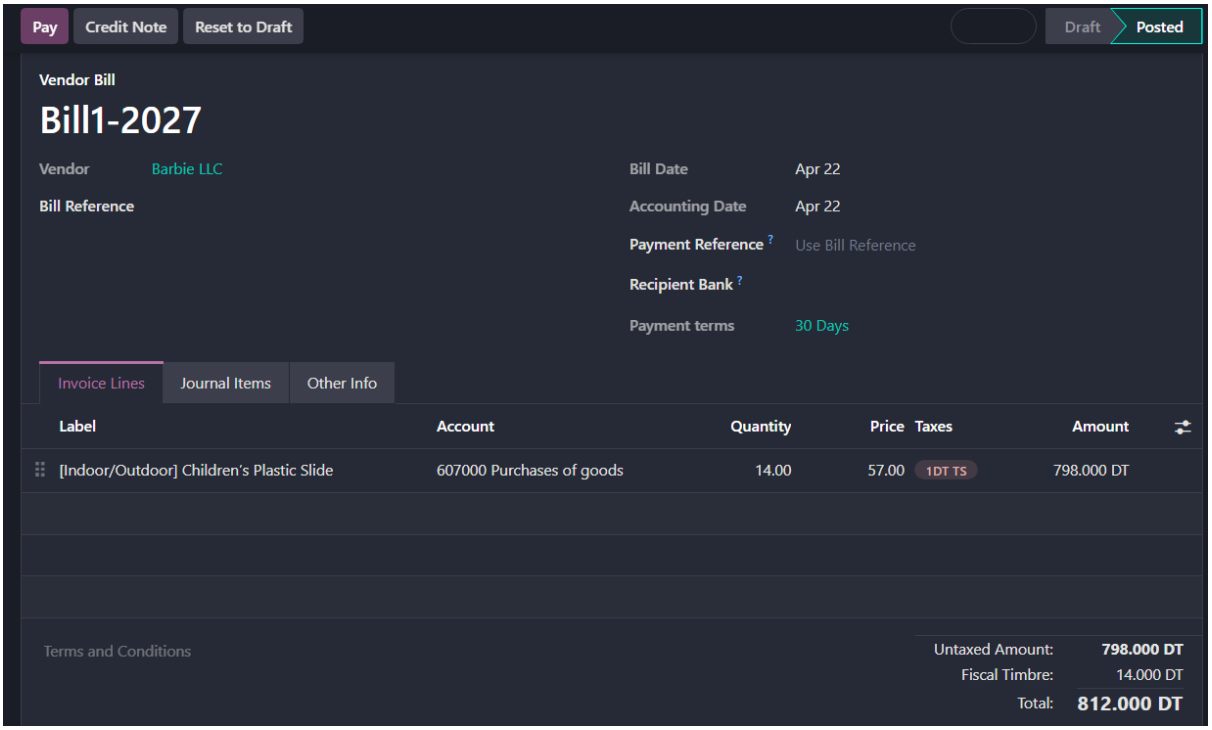
- Go back to the Purchase Order and click **Create Bill**.
- **Confirm** the bill to record it in accounting.
- Click **Register Payment** to complete the financial cycle.

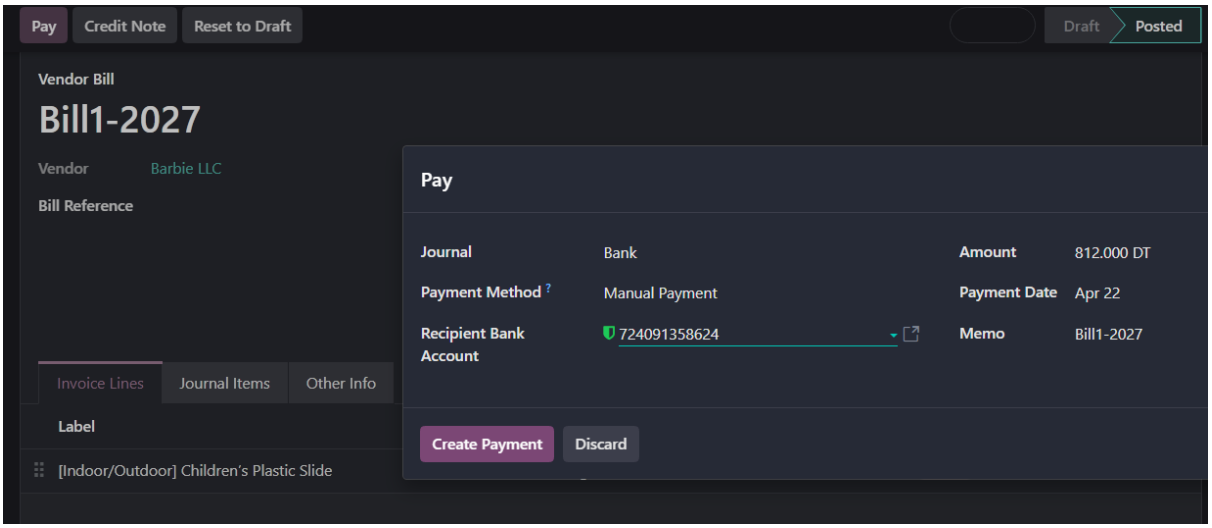


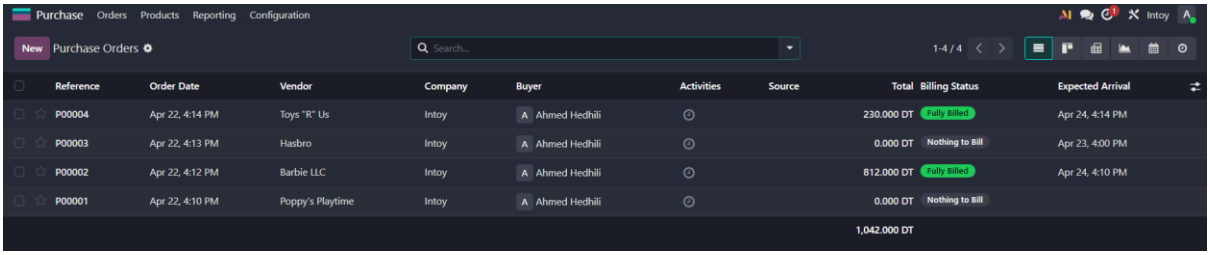
The second purchase

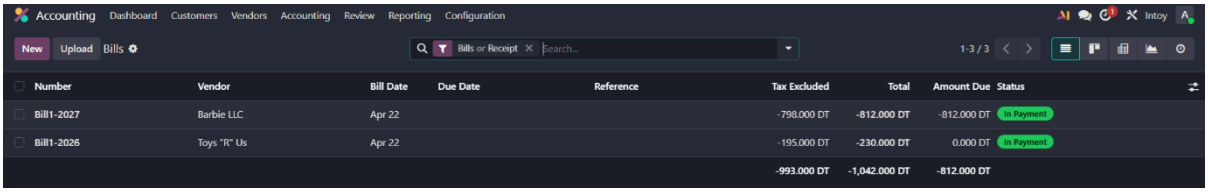


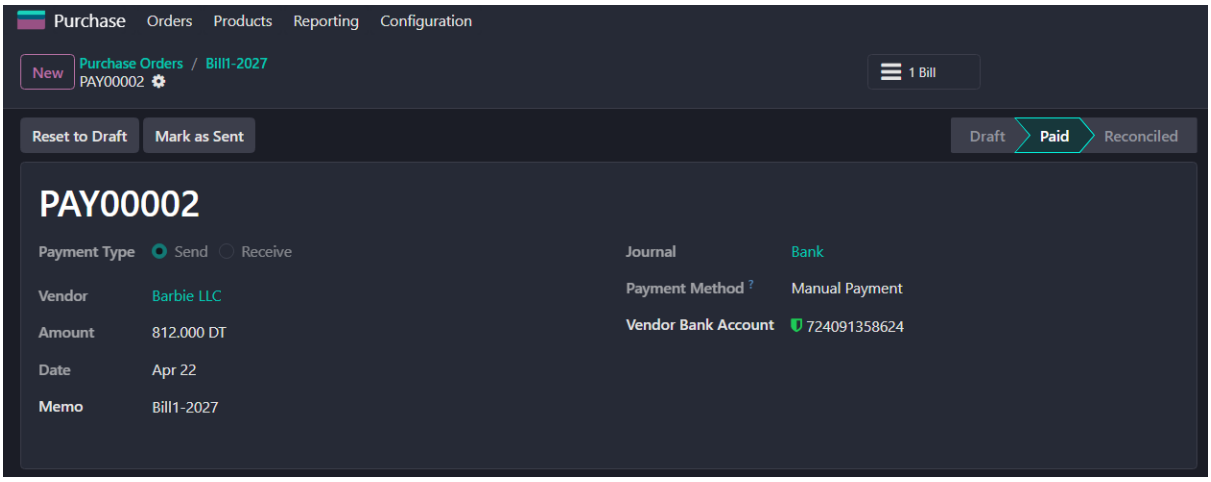

Purchase
Orders
Products
Reporting
Configuration
Intoy
New
Purchase Orders
1 selected
Create Bills
Cancel
Print
Actions
1-4 / 4
Reference
Order Date
Vendor
Company
Buyer
Activities
Source
Total
Billing Status
Expected Arrival
P00004 Apr 22, 4:14 PM Toys "R" Us Intoy Ahmed Hedhili 230.000 DT Fully Billed Apr 24, 4:14 PM
P00003 Apr 22, 4:13 PM Hasbro Intoy Ahmed Hedhili 0.000 DT Nothing to Bill Apr 23, 4:00 PM
P00002 Apr 22, 4:12 PM Barbie LLC Intoy Ahmed Hedhili 812.000 DT Waiting Bills Apr 24, 4:10 PM
P00001 Apr 22, 4:10 PM Poppy's Playtime Intoy Ahmed Hedhili 0.000 DT Nothing to Bill
812.000 DT


Pay
Credit Note
Reset to Draft
Draft
Posted
Vendor Bill
Bill1-2027
Vendor
Barbie LLC
Bill Reference
Bill Date
Apr 22
Accounting Date
Apr 22
Payment Reference
Use Bill Reference
Recipient Bank
Payment terms
30 Days
Invoice Lines
Journal Items
Other Info
Label
Account
Quantity
Price
Taxes
Amount
[Indoor/Outdoor] Children's Plastic Slide
607000 Purchases of goods
14.00
57.00
1DT TS
798.000 DT
Terms and Conditions
Untaxed Amount:
798.000 DT
Fiscal Timbre:
14.000 DT
Total:
812.000 DT


Pay
Credit Note
Reset to Draft
Draft
Posted
Vendor Bill
Bill1-2027
Vendor
Barbie LLC
Bill Reference
Pay
Journal
Bank
Amount
812.000 DT
Payment Method
Manual Payment
Payment Date
Apr 22
Recipient Bank Account
724091358624
Memo
Bill1-2027
Create Payment
Discard
Invoice Lines
Journal Items
Other Info
Label
[Indoor/Outdoor] Children's Plastic Slide



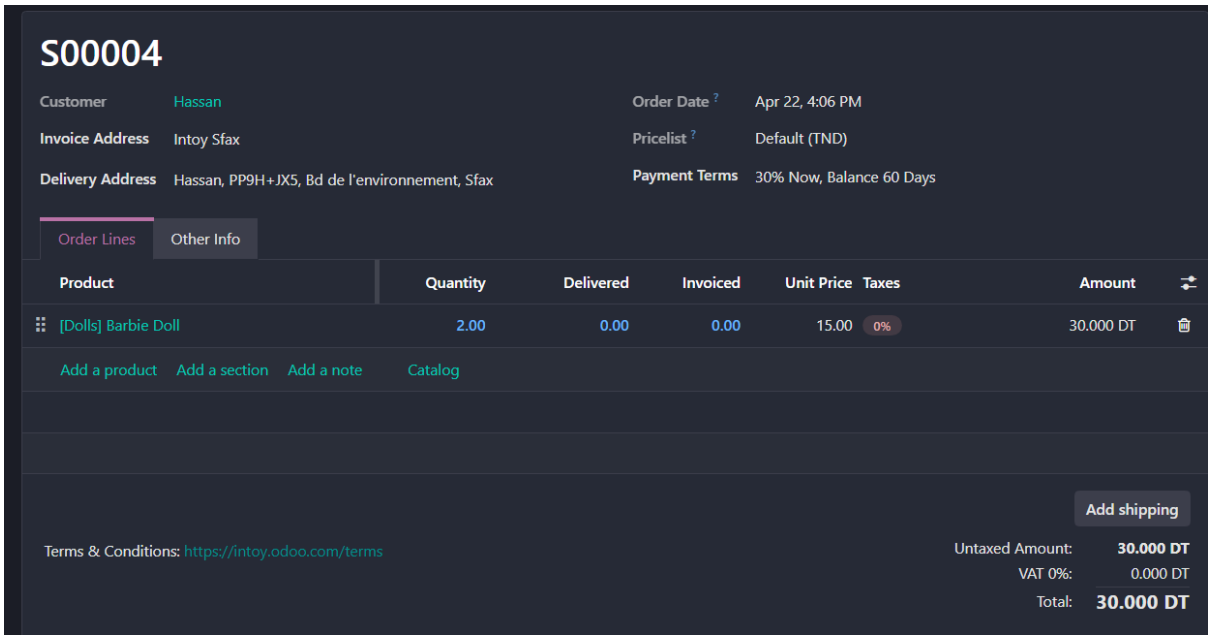


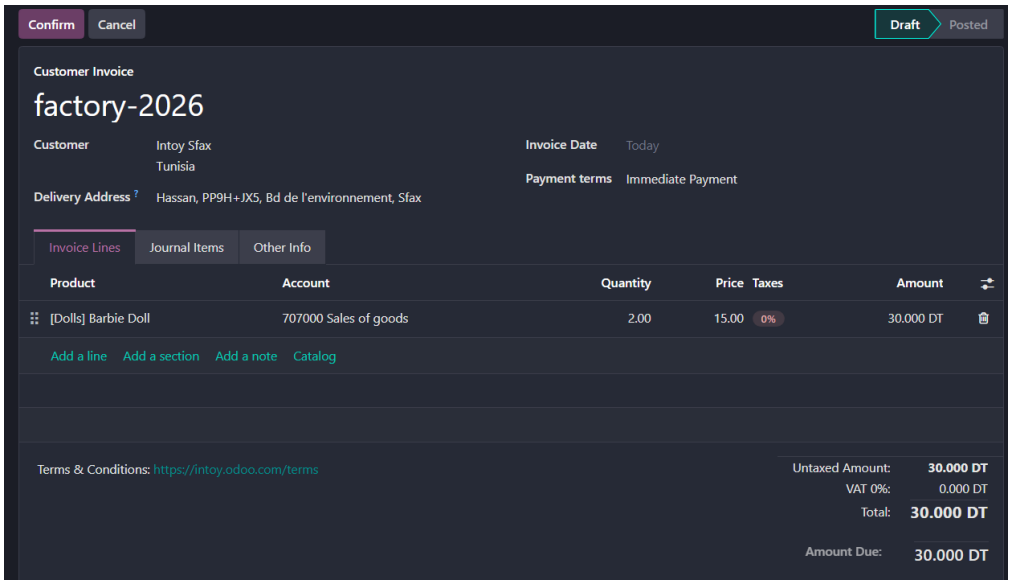


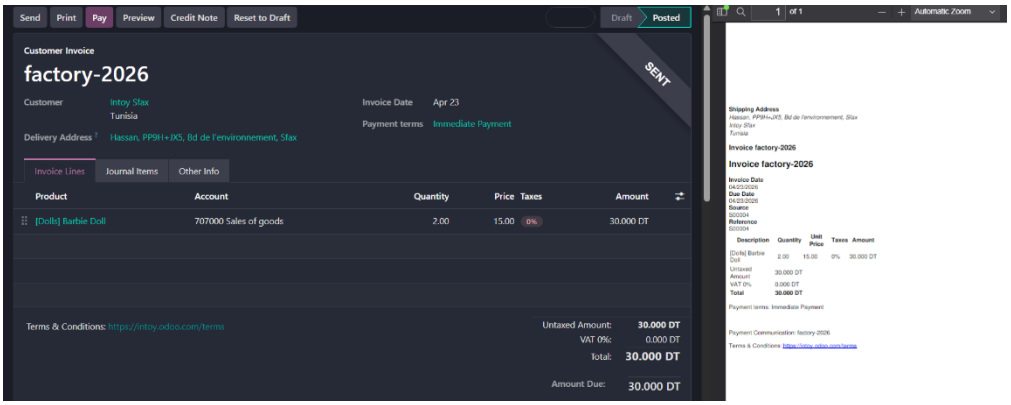
X. Activity 8: Sales & Inventory Audit

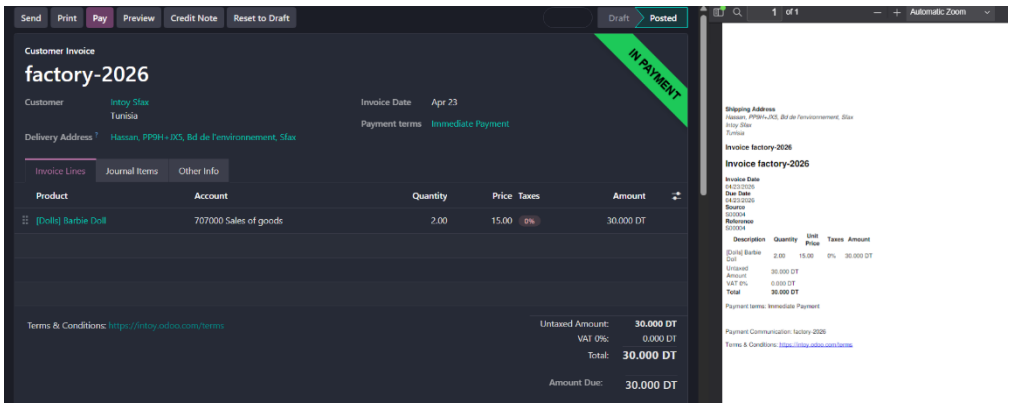
1- Sales Cycle

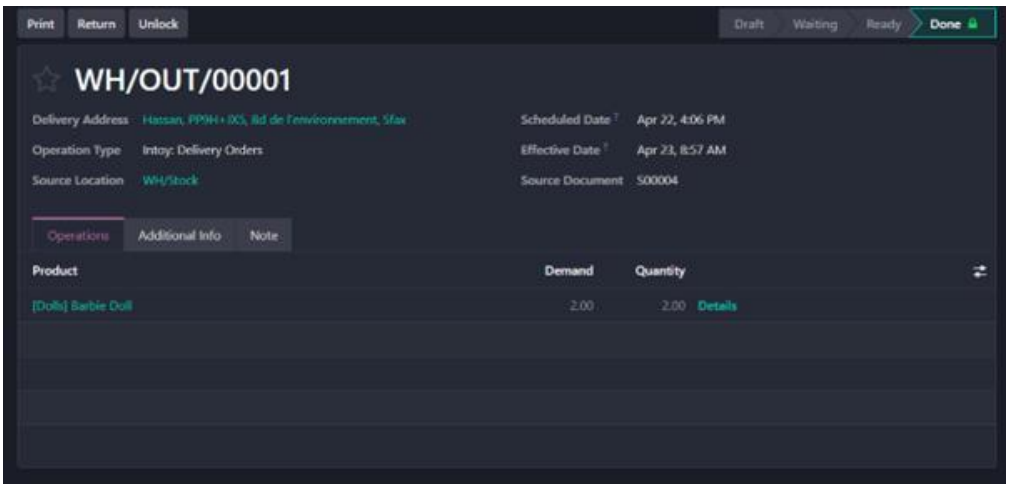
- The first order:



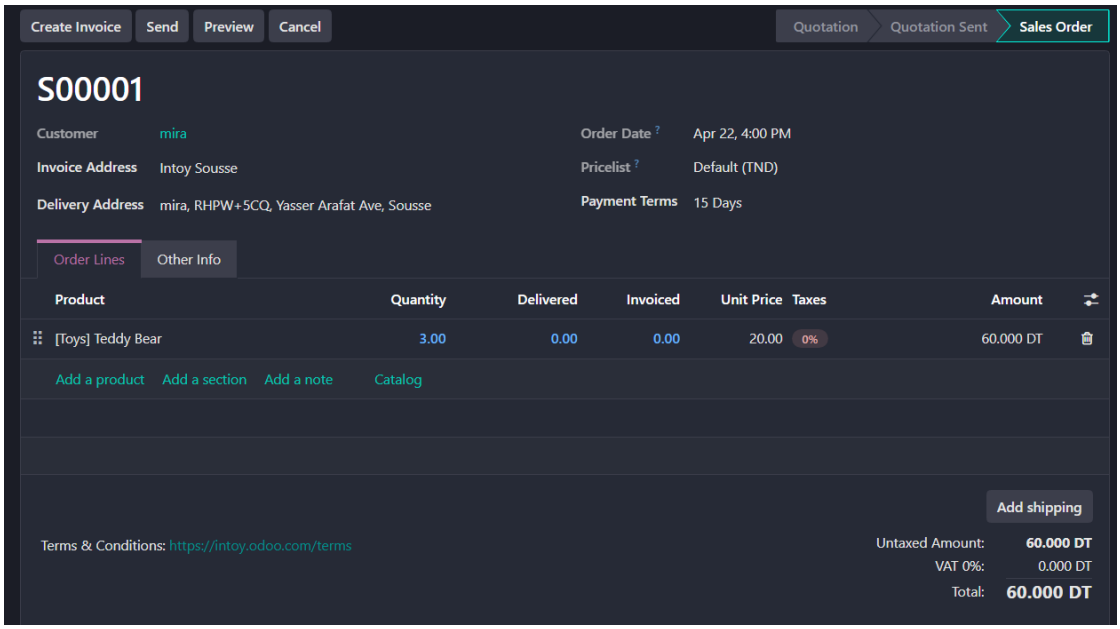

Confirm
Cancel
Draft
Posted
Customer Invoice
factory-2026
Customer
Intoy Sfax
Tunisia
Delivery Address
Hassan, PP9H+JX5, Bd de l'environnement, Sfax
Invoice Date
Today
Payment terms
Immediate Payment
Invoice Lines
Journal Items
Other Info
Product
Account
Quantity
Price
Taxes
Amount
[Dolls] Barbie Doll
707000 Sales of goods
2.00
15.00
0%
30.000 DT
Add a line
Add a section
Add a note
Catalog
Terms & Conditions: https://intoy.odoo.com/terms
Untaxed Amount:
30.000 DT
VAT 0%:
0.000 DT
Total:
30.000 DT
Amount Due:
30.000 DT

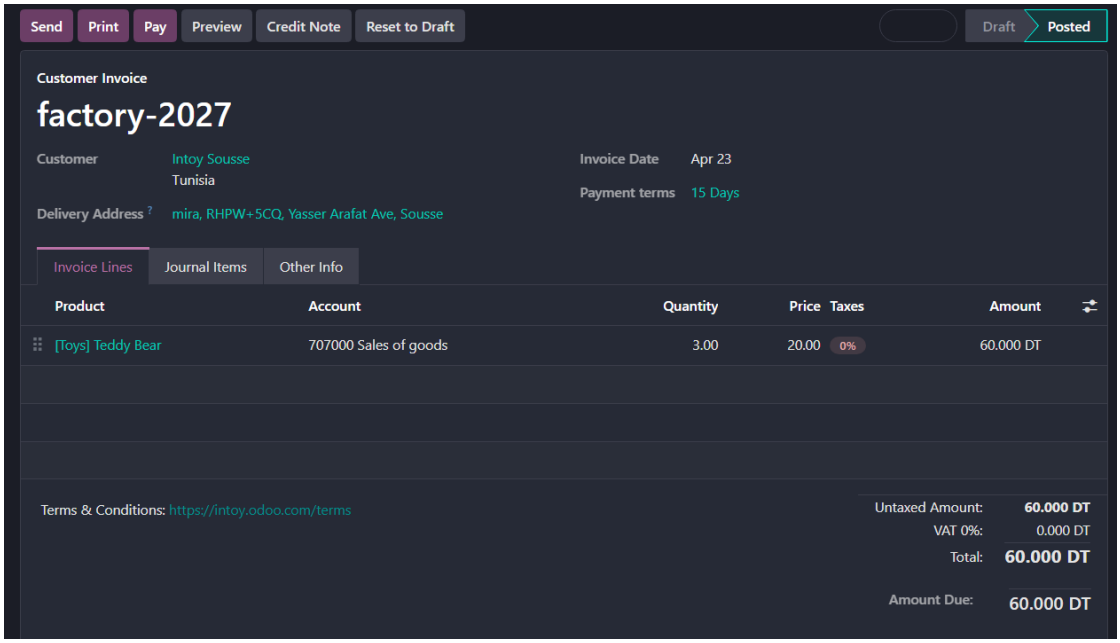

Send
Print
Pay
Preview
Credit Note
Reset to Draft
Draft
Posted
SENT
Customer Invoice
factory-2026
Customer
Intoy Sfax
Tunisia
Delivery Address
Hassan, PP9H+JX5, Bd de l'environnement, Sfax
Invoice Date
Apr 23
Payment terms
Immediate Payment
Invoice Lines
Journal Items
Other Info
Product
Account
Quantity
Price
Taxes
Amount
[Dolls] Barbie Doll
707000 Sales of goods
2.00
15.00
0%
30.000 DT
Terms & Conditions: https://intoy.odoo.com/terms
Untaxed Amount:
30.000 DT
VAT 0%:
0.000 DT
Total:
30.000 DT
Amount Due:
30.000 DT
Automatic Zoom
Invoice factory-2026

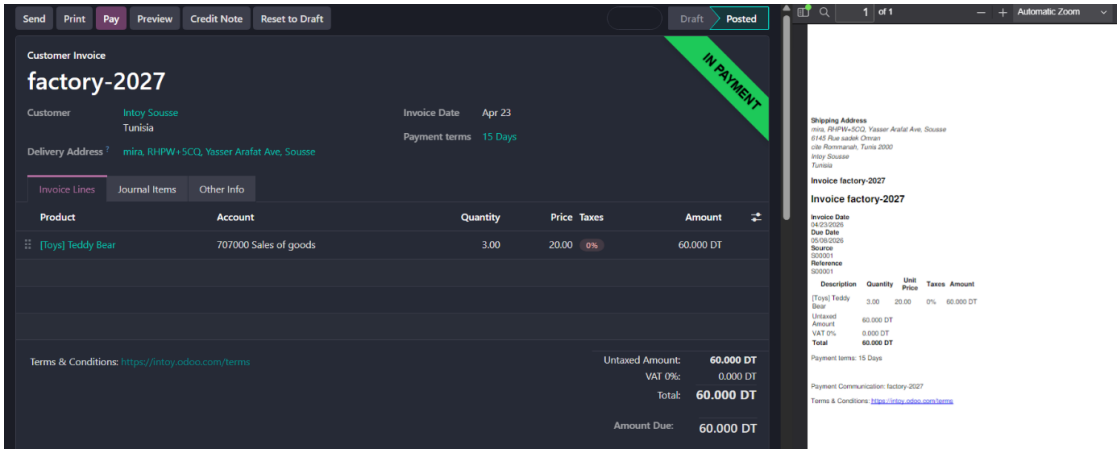

Send
Print
Pay
Preview
Credit Note
Reset to Draft
Draft
Posted
IN PAYMENT
Customer Invoice
factory-2026
Customer
Intoy Sfax
Tunisia
Delivery Address
Hassan, PP9H+JX5, Bd de l'environnement, Sfax
Invoice Date
Apr 23
Payment terms
Immediate Payment
Invoice Lines
Journal Items
Other Info
Product
Account
Quantity
Price
Taxes
Amount
[Dolls] Barbie Doll
707000 Sales of goods
2.00
15.00
0%
30.000 DT
Terms & Conditions: https://intoy.odoo.com/terms
Untaxed Amount:
30.000 DT
VAT 0%:
0.000 DT
Total:
30.000 DT
Amount Due:
30.000 DT
Automatic Zoom
Invoice factory-2026

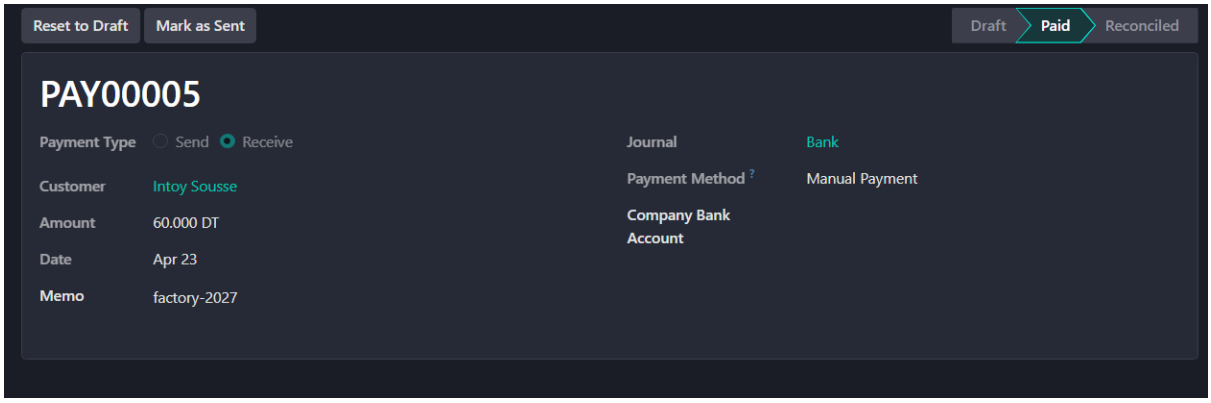

Print
Return
Unlock
Draft
Waiting
Ready
Done
WH/OUT/00001
Delivery Address
Hassan, PP9H+JX5, Bd de l'environnement, Sfax
Operation Type
Intoy: Delivery Orders
Source Location
WH/Stock
Scheduled Date
Apr 22, 4:06 PM
Effective Date
Apr 23, 8:57 AM
Source Document
S00004
Operations
Additional Info
Note
Product
Demand
Quantity
[Dolls] Barbie Doll
2.00
2.00
Details

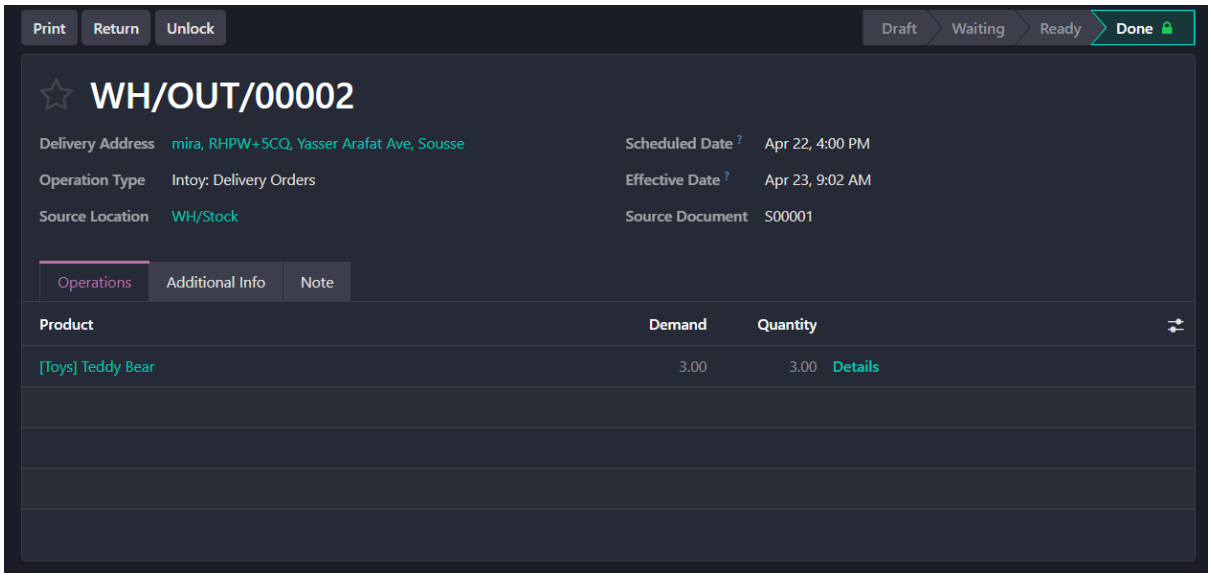
The second order

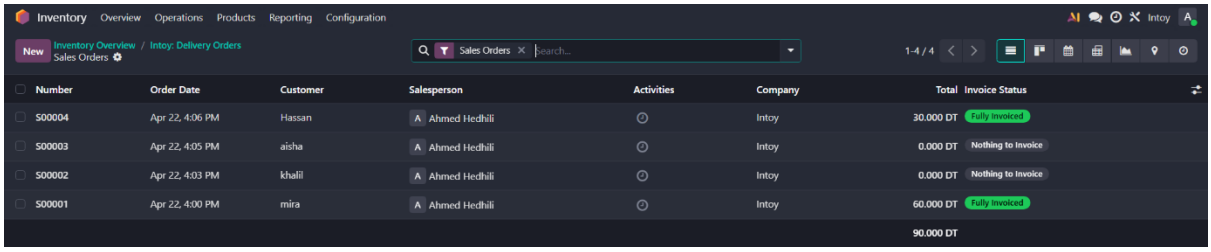






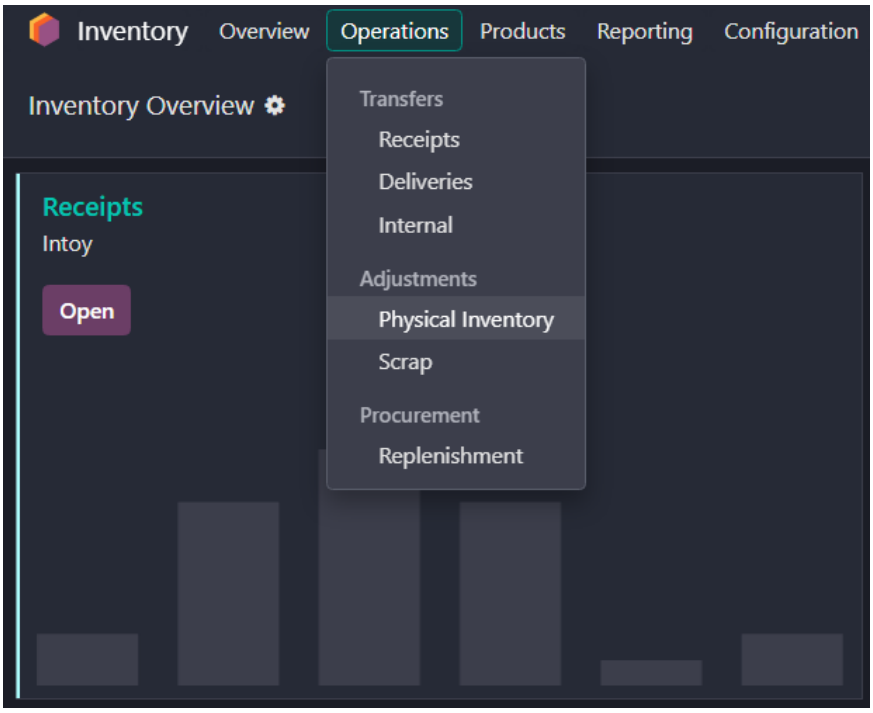






2- Inventory Audit/Physical Inventory

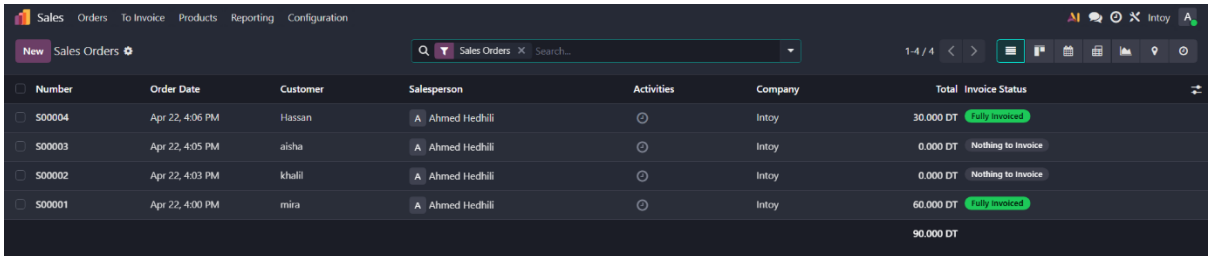
The goal is to ensure that the quantities on the shelves match what is recorded in Odoo.

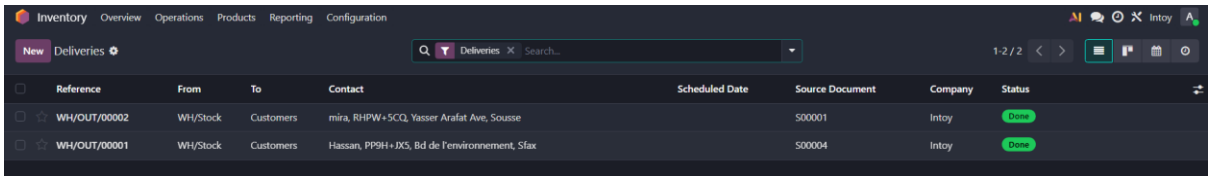


XI. Activity 9: Payment Collection

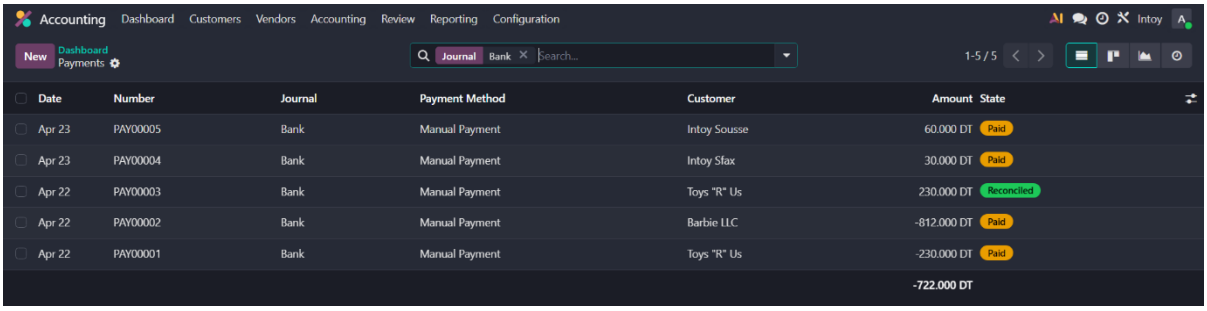
We will focus on completing the payment collection process in accounting section.

1- Sales & Collection

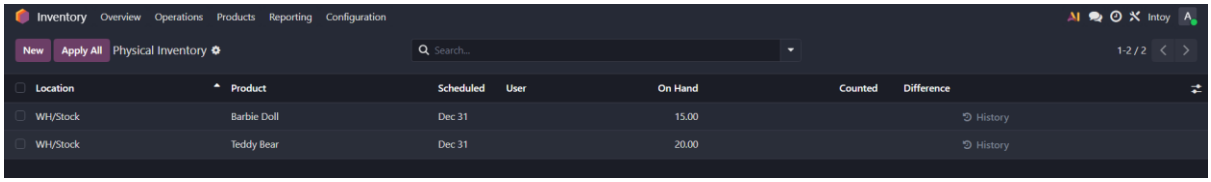




2- Reconciliation

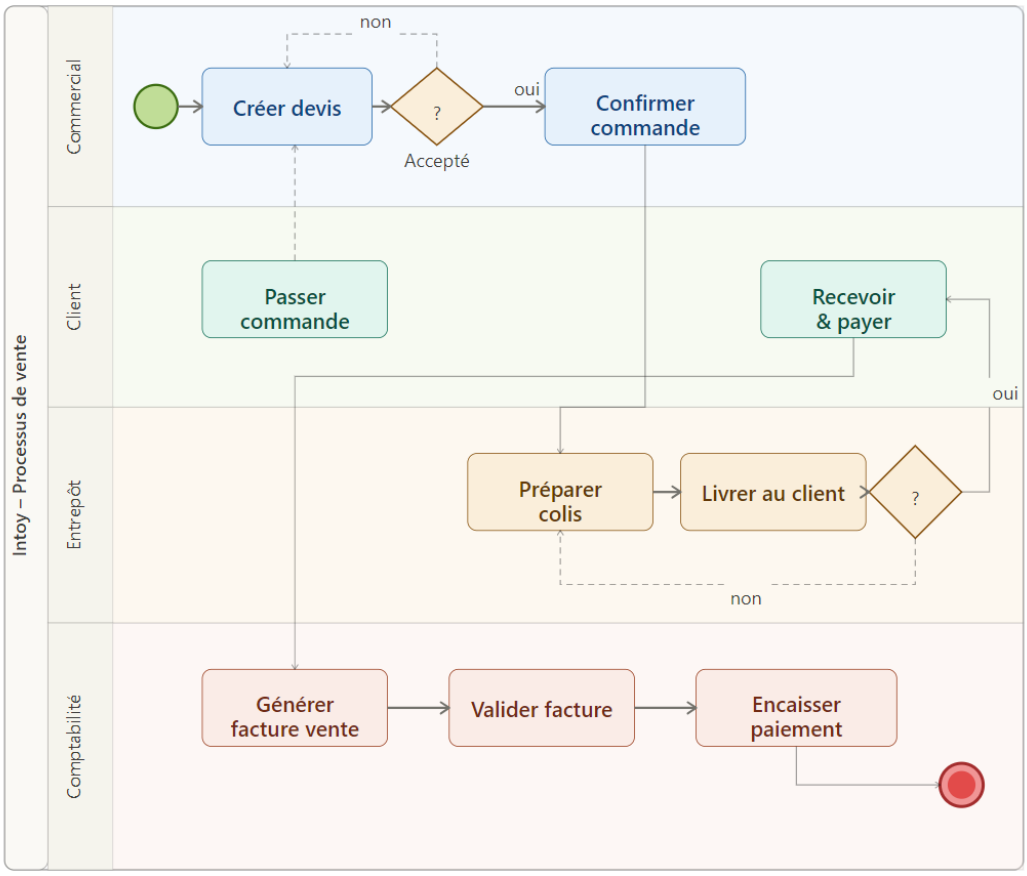


3- Inventory Audit

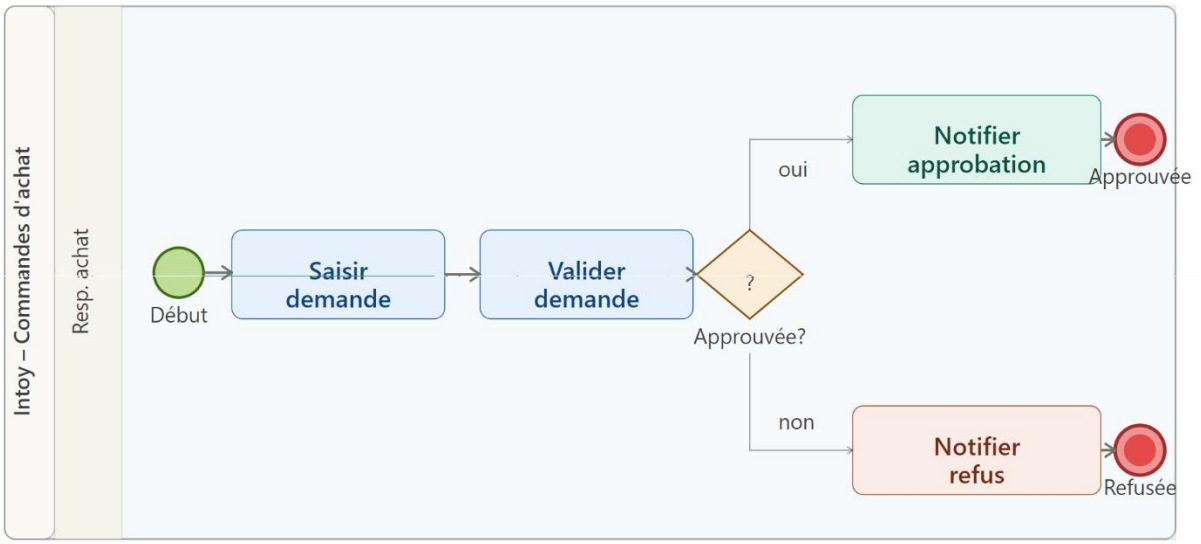


XII. Conception with 2 models BPMN:

1) Sales Process:



2) Purchase approval:



XIII. Conclusion

This project provided a comprehensive practical experience in implementing an ERP system using the **Odoo** platform for our toy shop, **Intoy**. By integrating core business modules—CRM, Sales, Purchases, Inventory, and Accounting—we successfully simulated a complete business lifecycle. We learned how to manage master data, automate workflows from procurement to payment collection, and ensure data consistency across departments. This implementation demonstrates how ERP systems enhance operational efficiency, reduce human error, and provide real-time insights for better decision-making in modern business environments.